

Age-depth Modelling

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Schedule

Today:

Lectures – feel free to interrupt with questions

Tomorrow:

- Short recap, questions**
- Exercises and discussion, wrap-up**

Learning objectives

You will have a broad overview of a **wide range** of commonly-applied age-depth modelling **techniques**, their **advantages** and disadvantages, as well as dangerous **pitfalls**.

You will have the knowledge and practical background to defend your choice of age-depth models to co-authors and reviewers.

Richard Telford et al., (2004): All age models are wrong: but how badly?

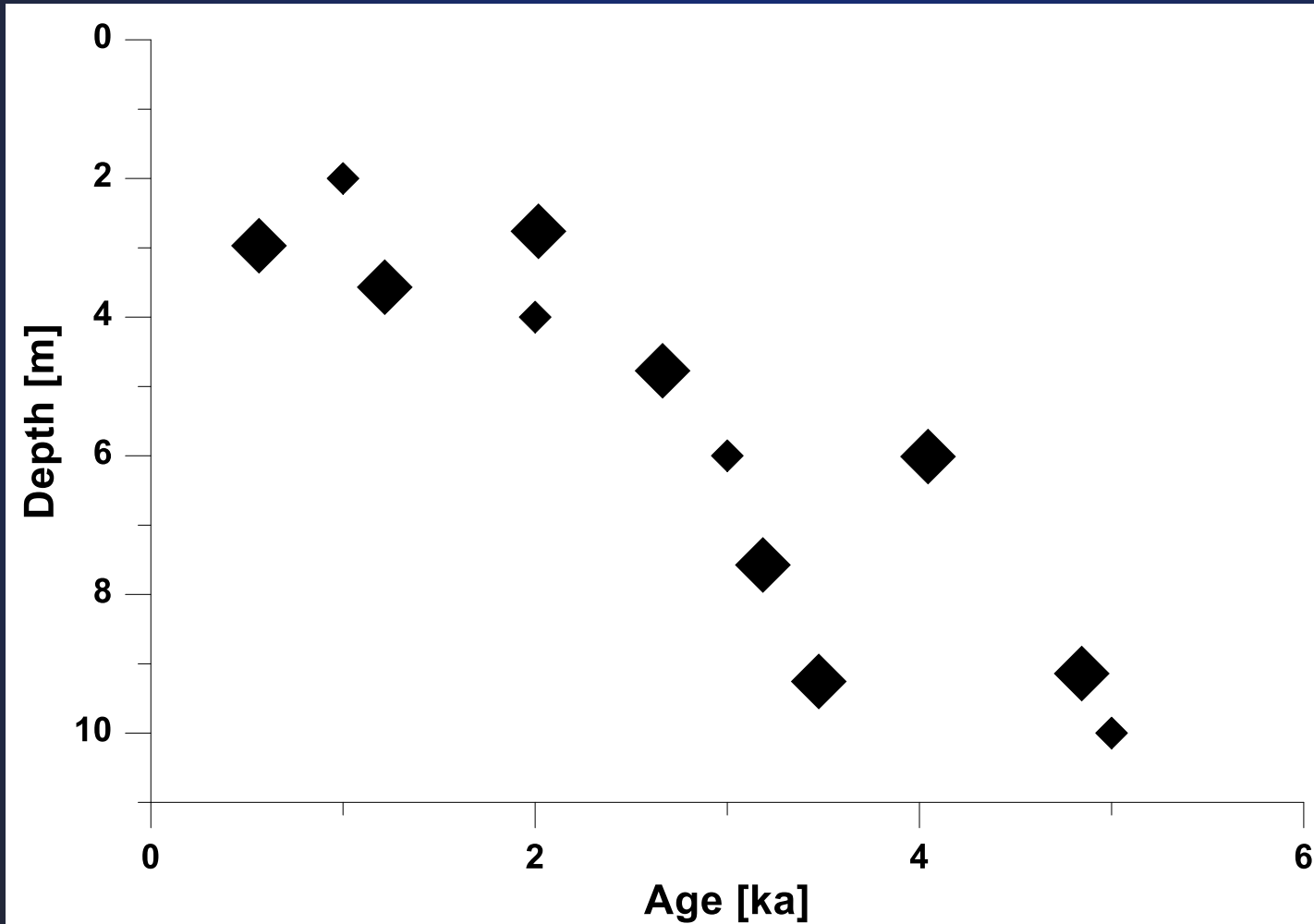
History

Why are models needed?

Dates: Scatter, outliers – how deal with these?

Questions: What's the nature of sedimentation: constant, episodically, pulses, hiatuses

History



History

- Should we try to fit a line through all dates?
- What else?
- Study design/number of dates required for answering a question? More is better or more is more problems?

¹⁴C models early 1990s (Bayesian)

- Specific calculation methods due to limited computing power
- Some sedimentation rate based

What kind of information goes into age-depth models?

The focus in this short-course will be on:

^{14}C dating

Luminescence (OSL)

U-Pb dating

But every technique has its specific problems.

What kind of information goes into age-depth models?

We will also touch upon techniques to include duration estimates (*"rates"*, *"relative time"*) into age-depth models.



Milankovitch
cyclostratigraphy

Varve layers

Tree rings

Annual ice layers

What kind of information goes into age-depth models?

When age models are constructed, the focus often lies on *dates*, not so much on *rates* (durations).

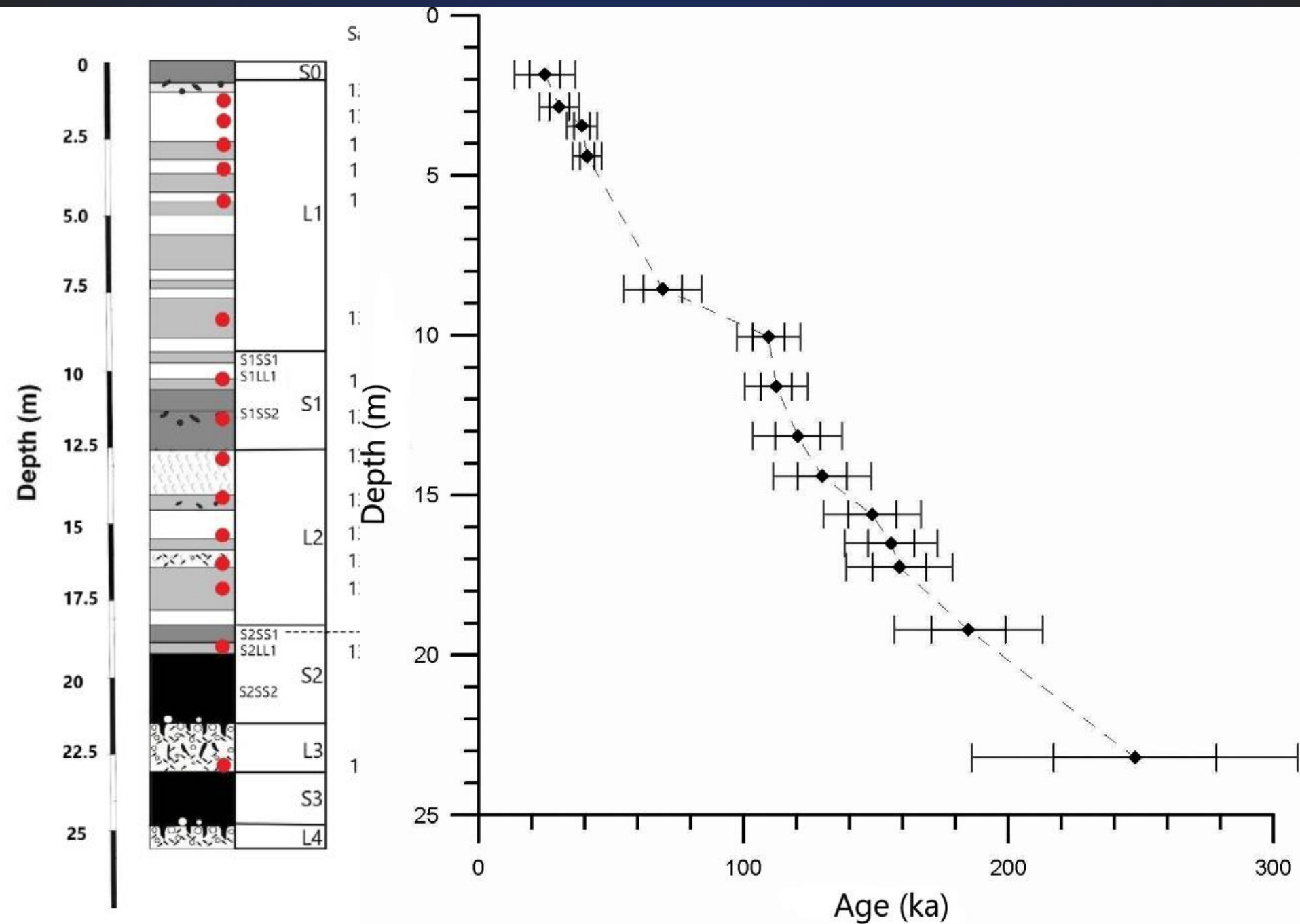
Next few slides: focus on dates.

Later: How can we integrate information from dates and rates?

What kind of age models are there?

Basic age-depth models – basically, you can create those in Excel.

Bayesian age-depth models – These age models use chronologic information from *dates* + some extra assumptions. For example, these age models explicitly incorporate assumptions on stratigraphy and/or sedimentation rate. However, Bayesian do not necessarily incorporate information on *rates*.



Dating Uncertainties

- Random vs. systematic uncertainties

Dating Uncertainties

Distribution of uncertainty? Gaussian?

Long tails?

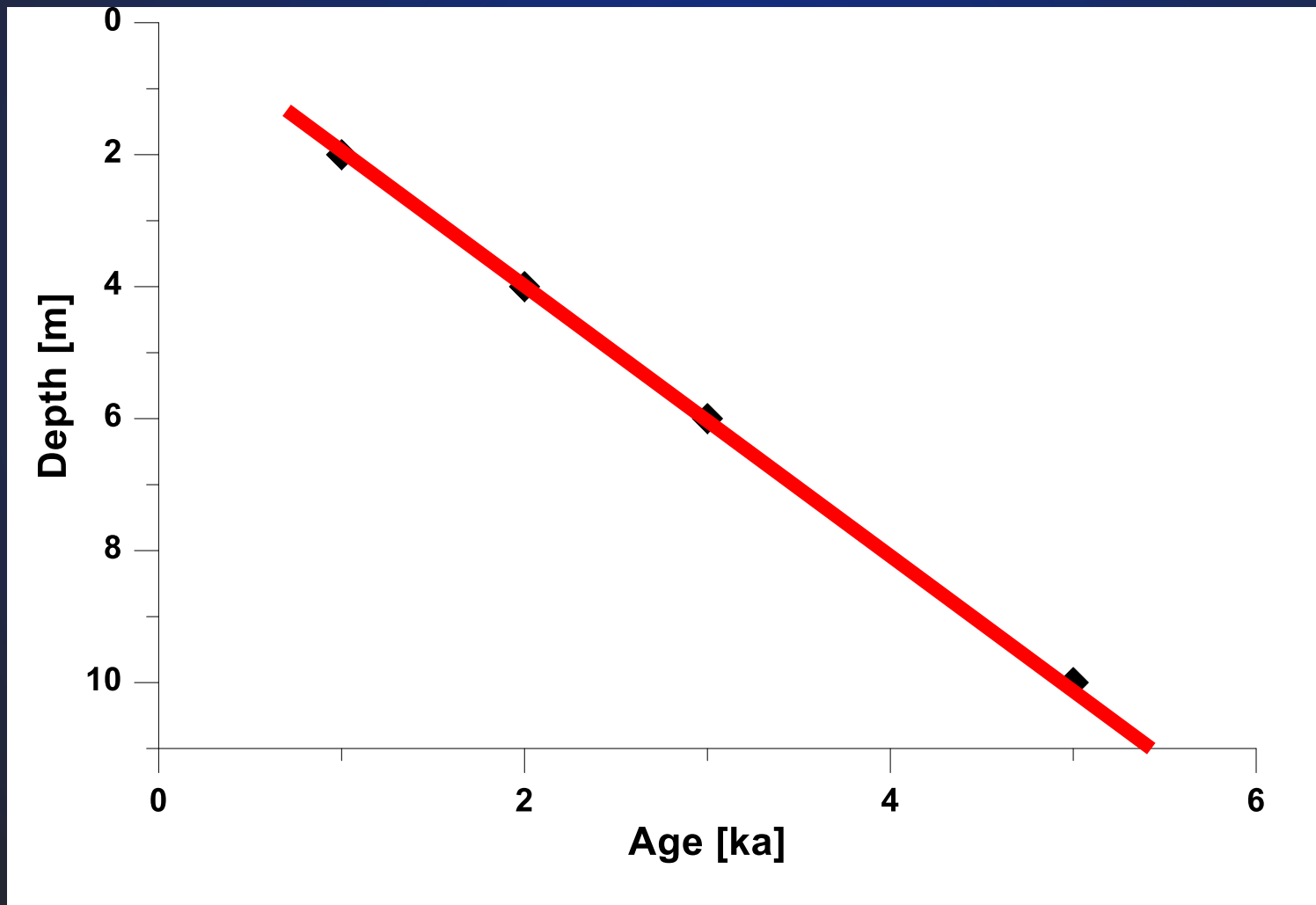
Is uncertainty realistic?

We know age (mean age)

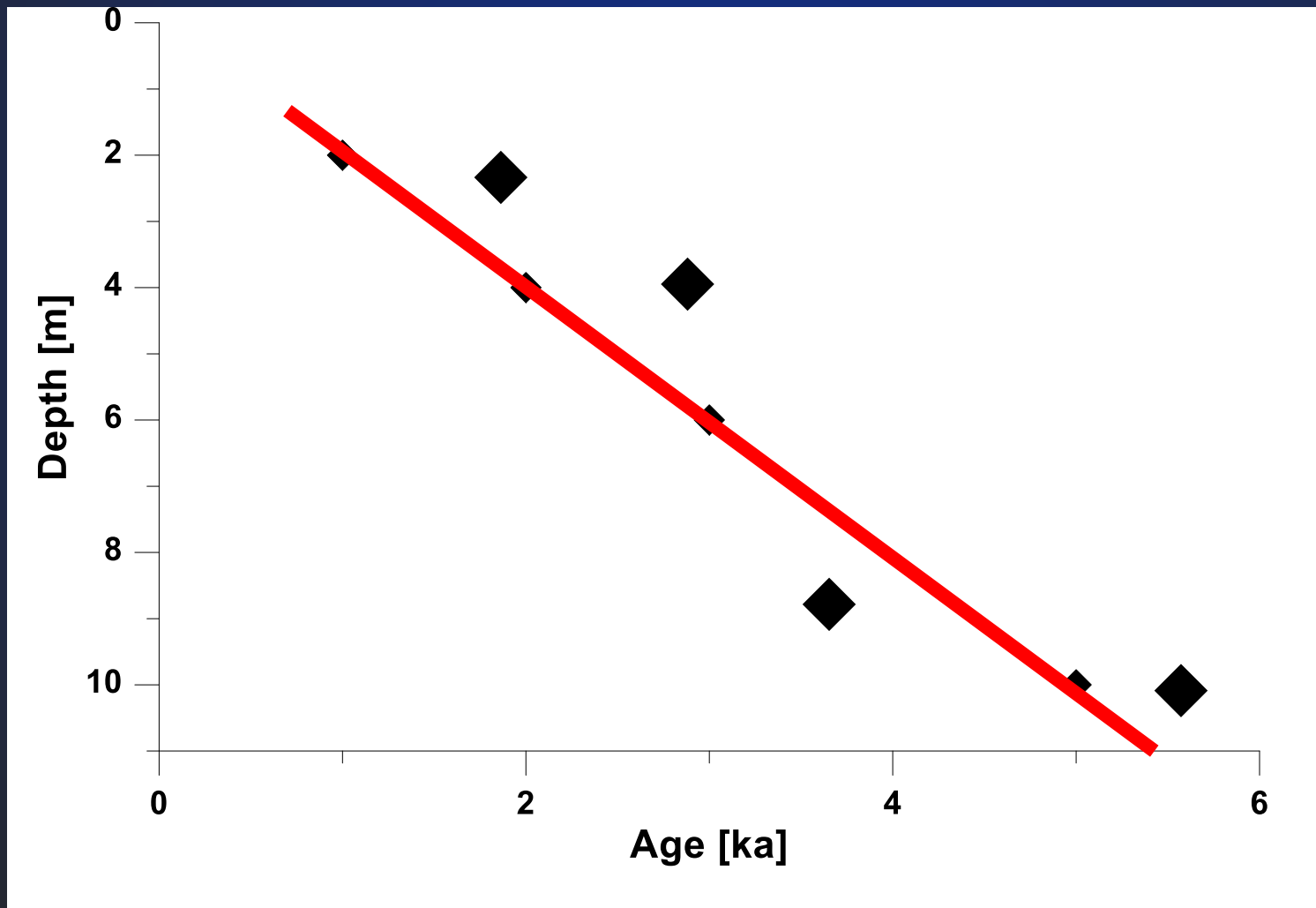
How good do we know uncertainty?

^{14}C calibration 'wiggles matching':
After calibration ^{14}C data are not normally
distributed, but often have (very) complex
uncertainty structures

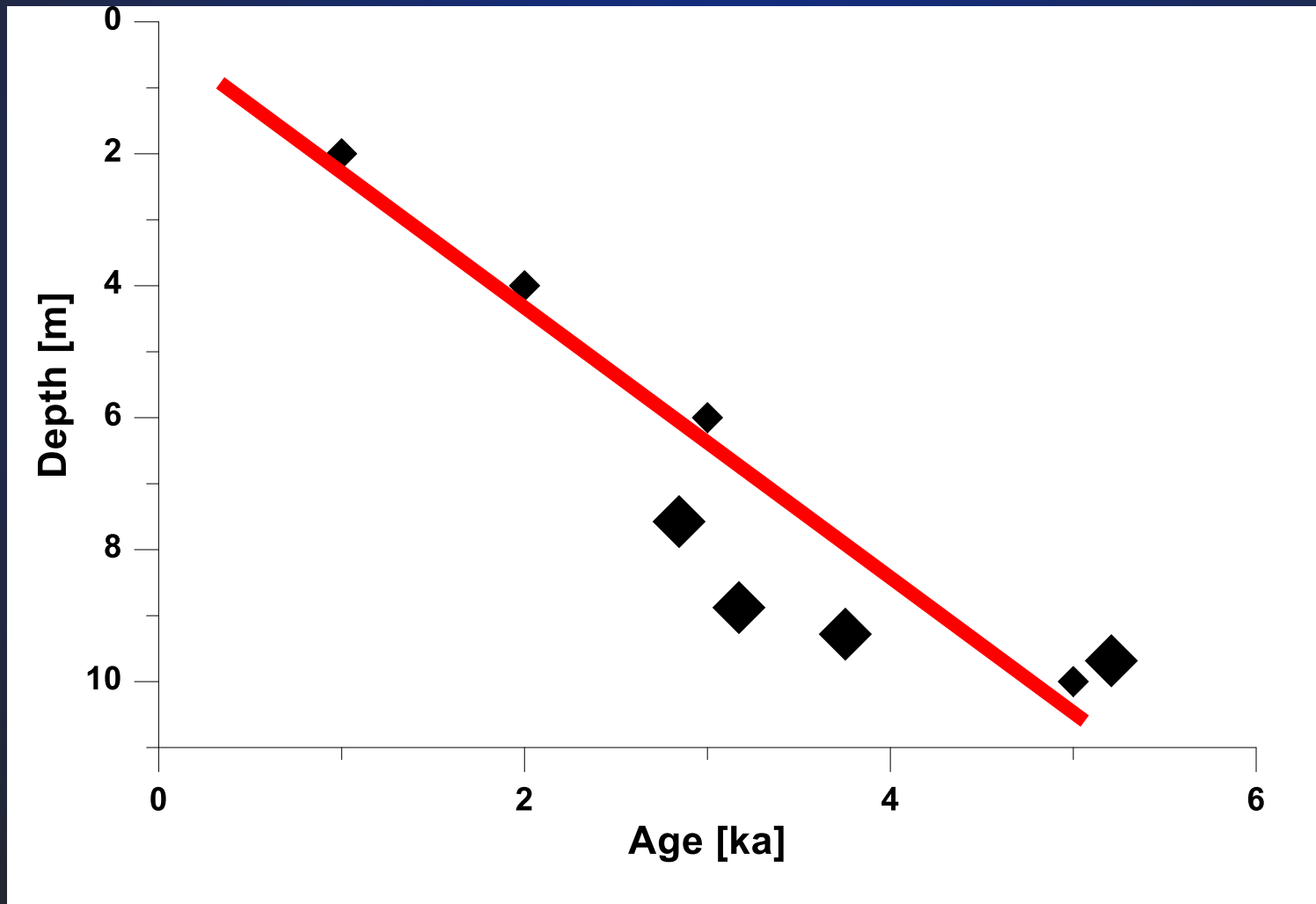
Age-depth modelling: linear regression



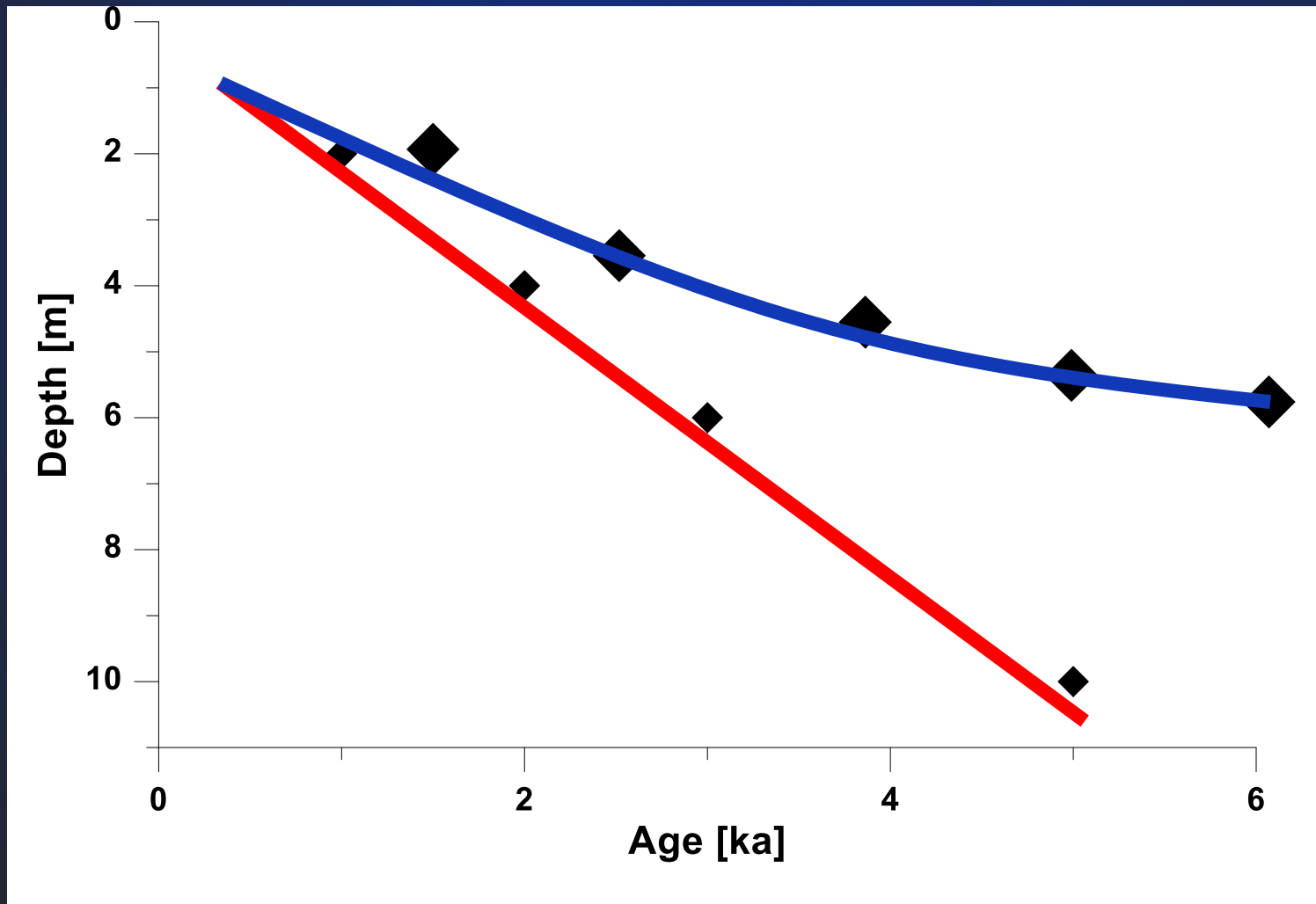
Age-depth modelling: linear regression



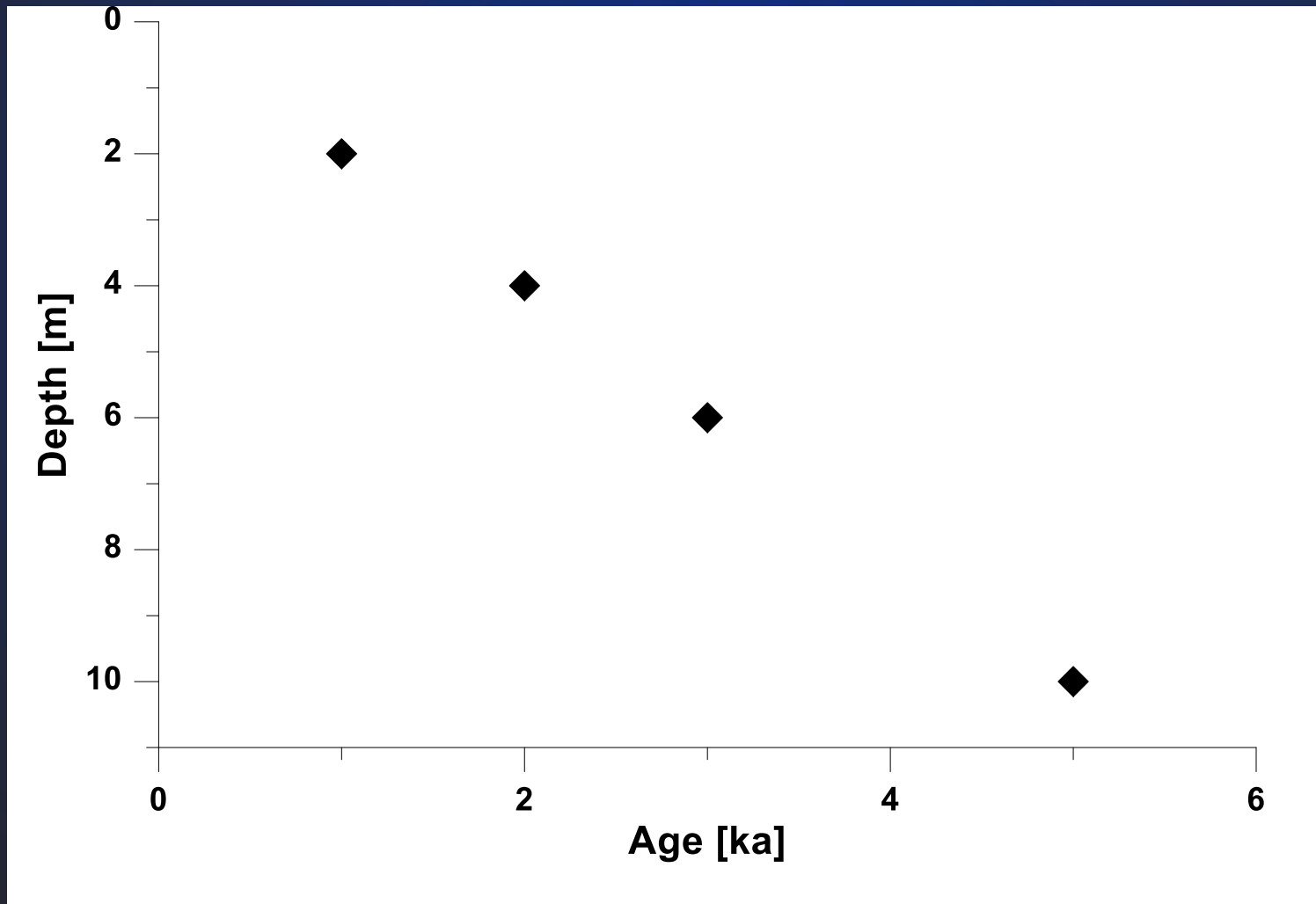
Age-depth modelling: how?



Age-depth modelling: how to interpret ages



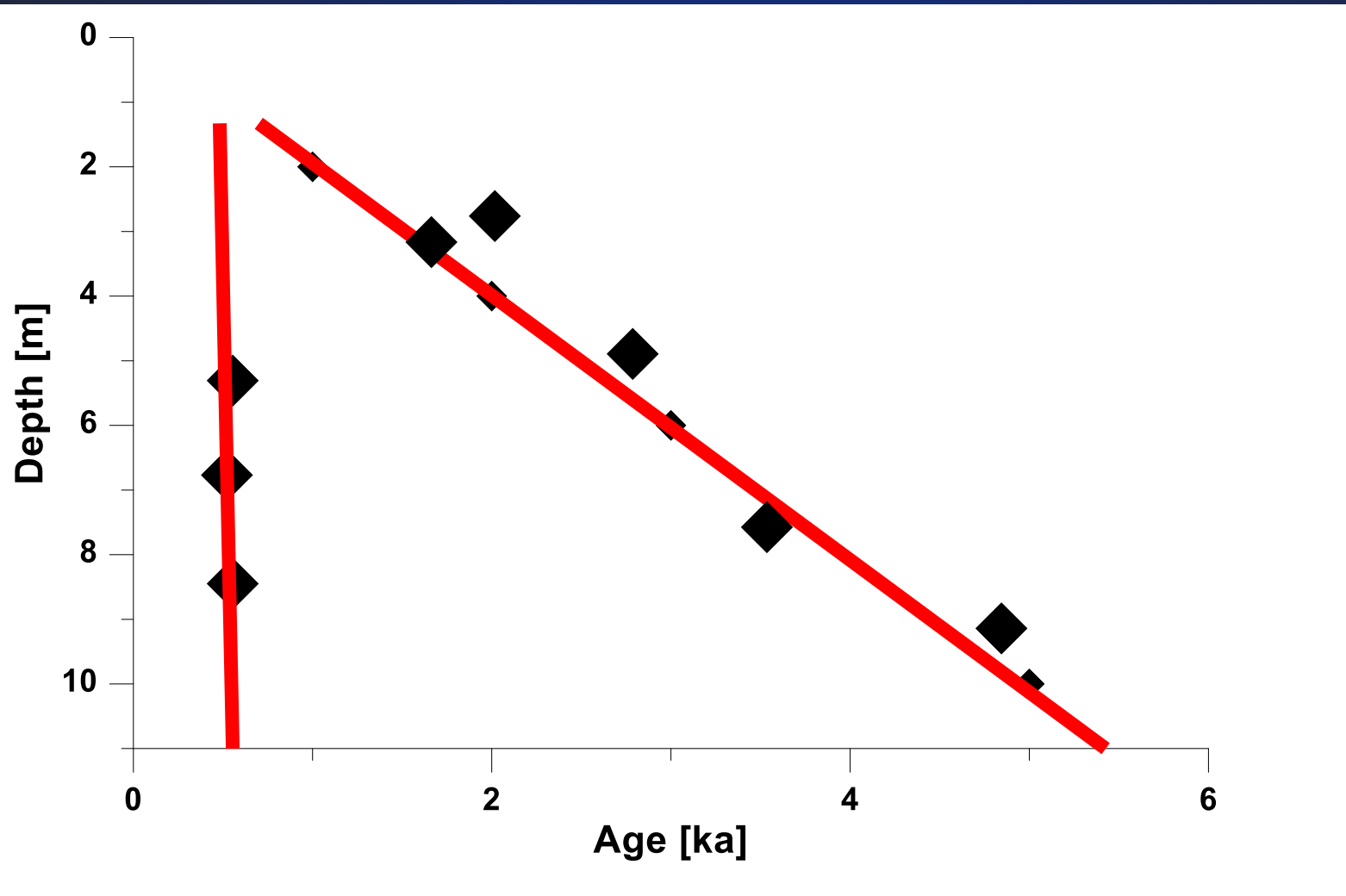
Age-depth modelling: how to interpret ages



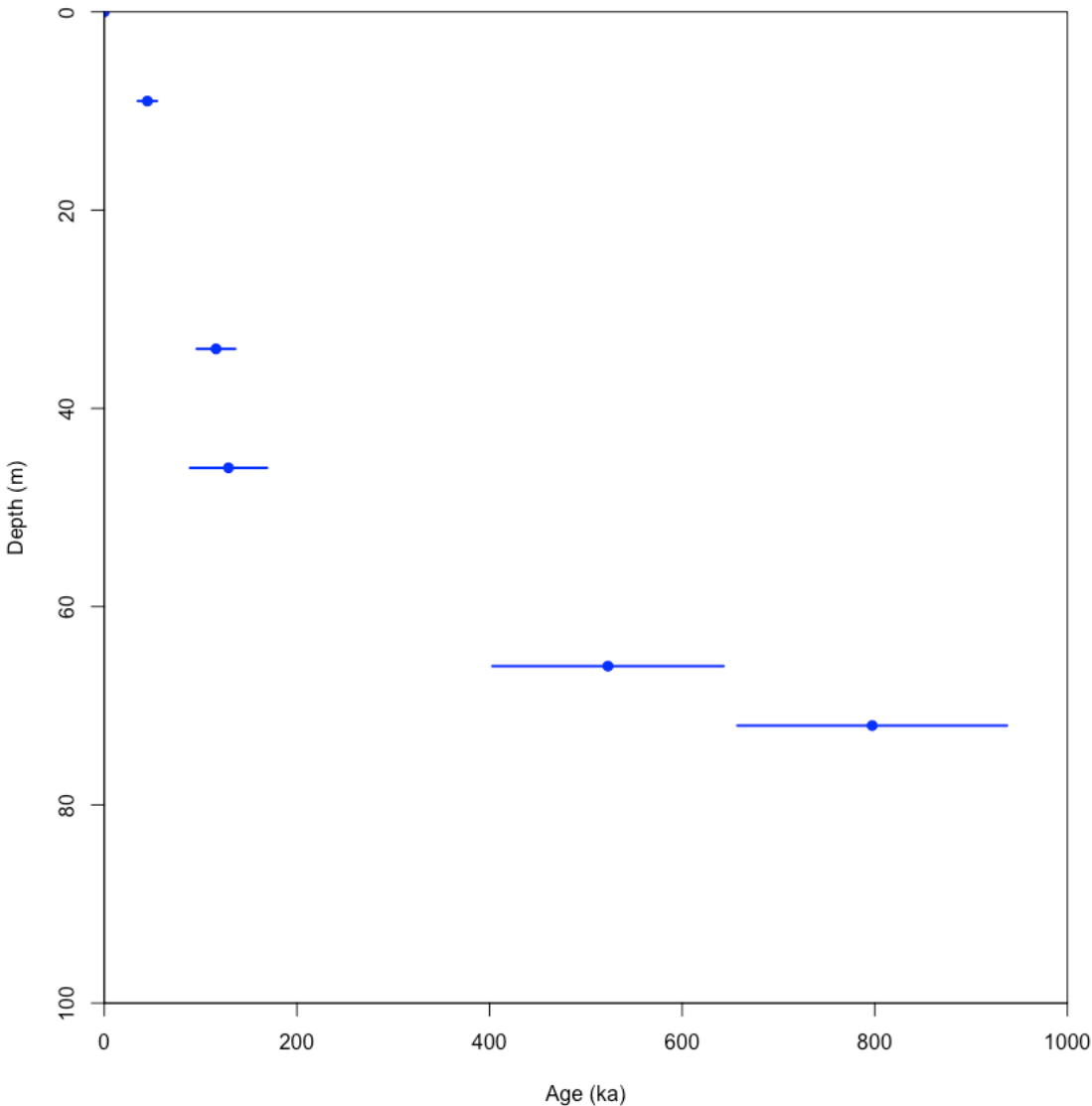
Summary:

- Outliers exist
- Uncertainty is often optimistic and accounts for only analytical uncertainty (not: bioturbation, re-deposition, etc.)
- Sometimes multiple interpretations are possible
- Understanding uncertainty is important
- A Geoscientific interpretation is important

Age-depth modelling: how to interpret ages?



Which age-depth model?



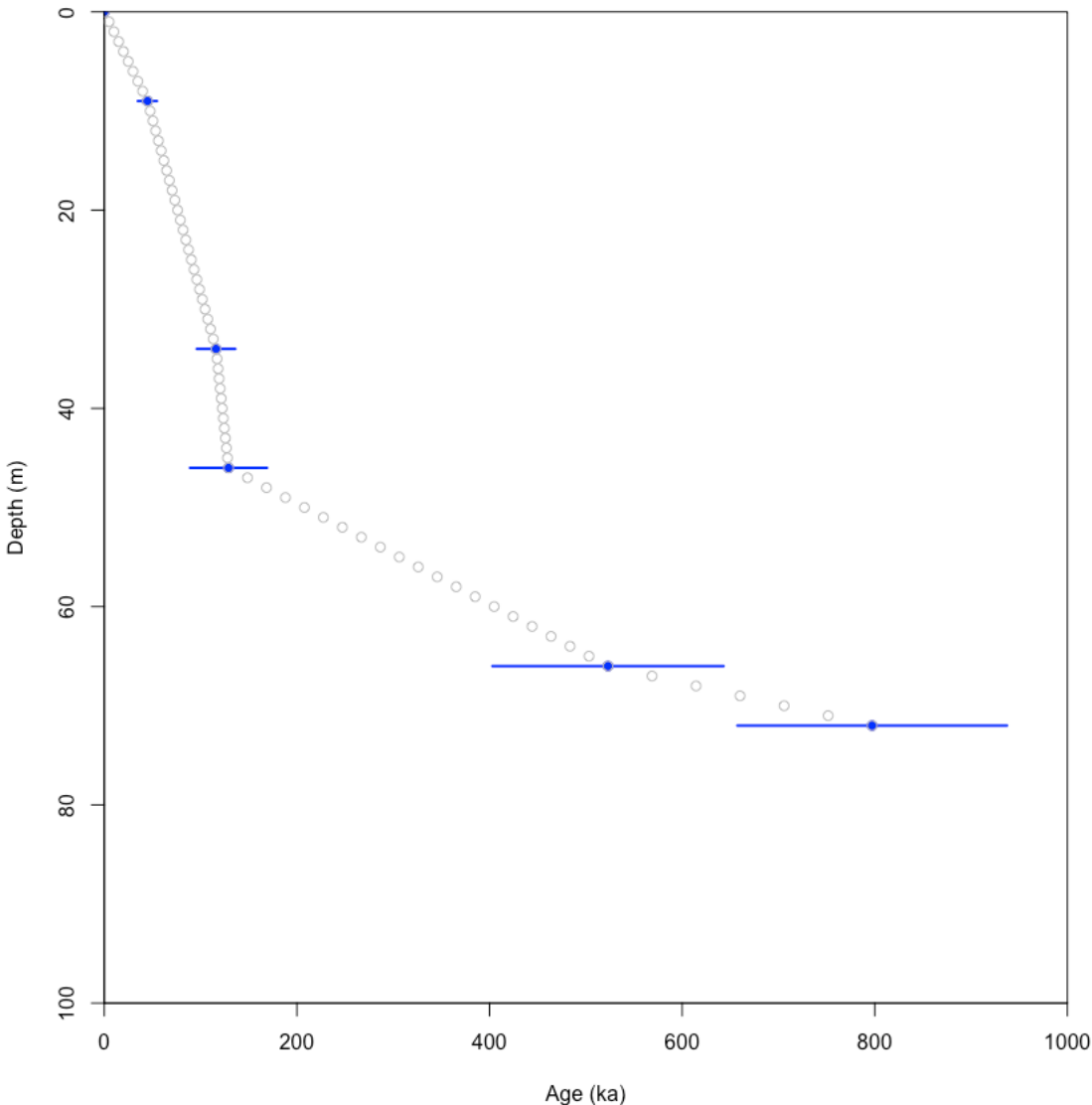
Lake GLOMAR

100 m core

5 radio-isotopic dates

surface = recent

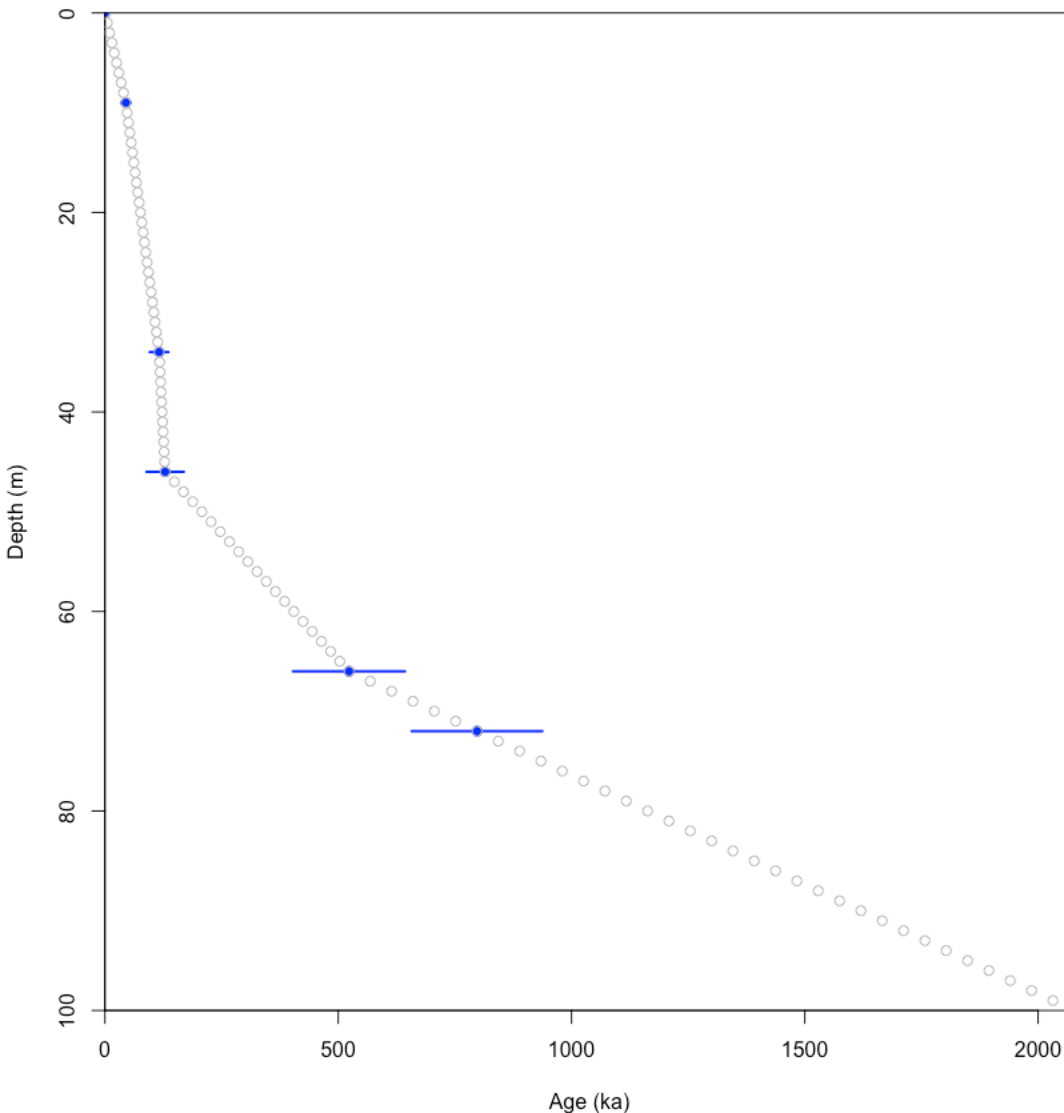
Linear interpolation might put too much weight on individual dates and cause strong fluctuations in sed. rate.



Linear interpolation

Too much weight on individual dates?

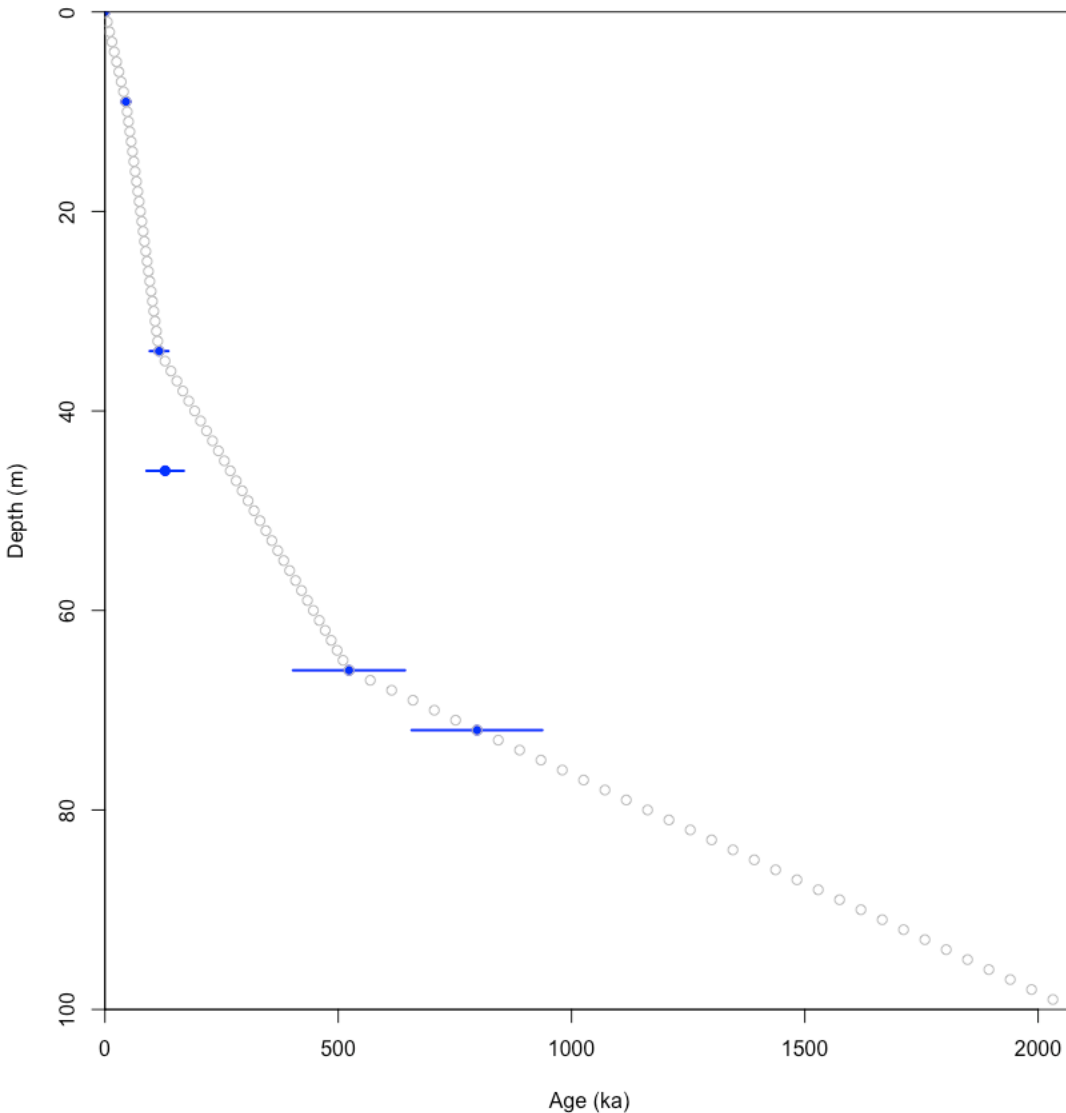
Linear interpolation might put too much weight on individual dates and cause strong fluctuations in sedimentation rate.



**Linear interpolation
and extrapolation**

Too much weight on
individual dates?

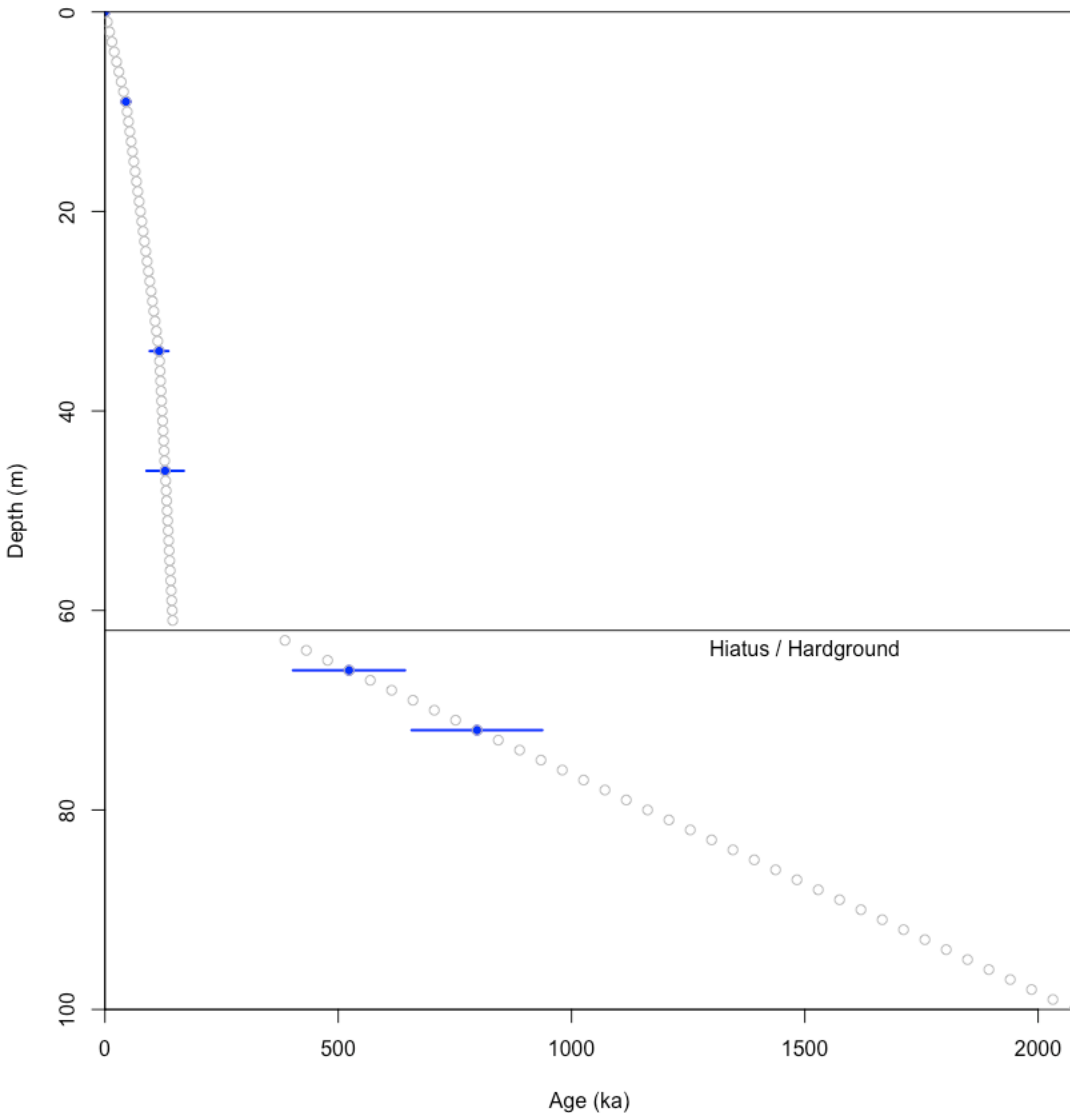
Removing date 3 results in a smoother evolution of sedimentation rates...



**Linear interpolation
and extrapolation**

→ Date 3 = outlier

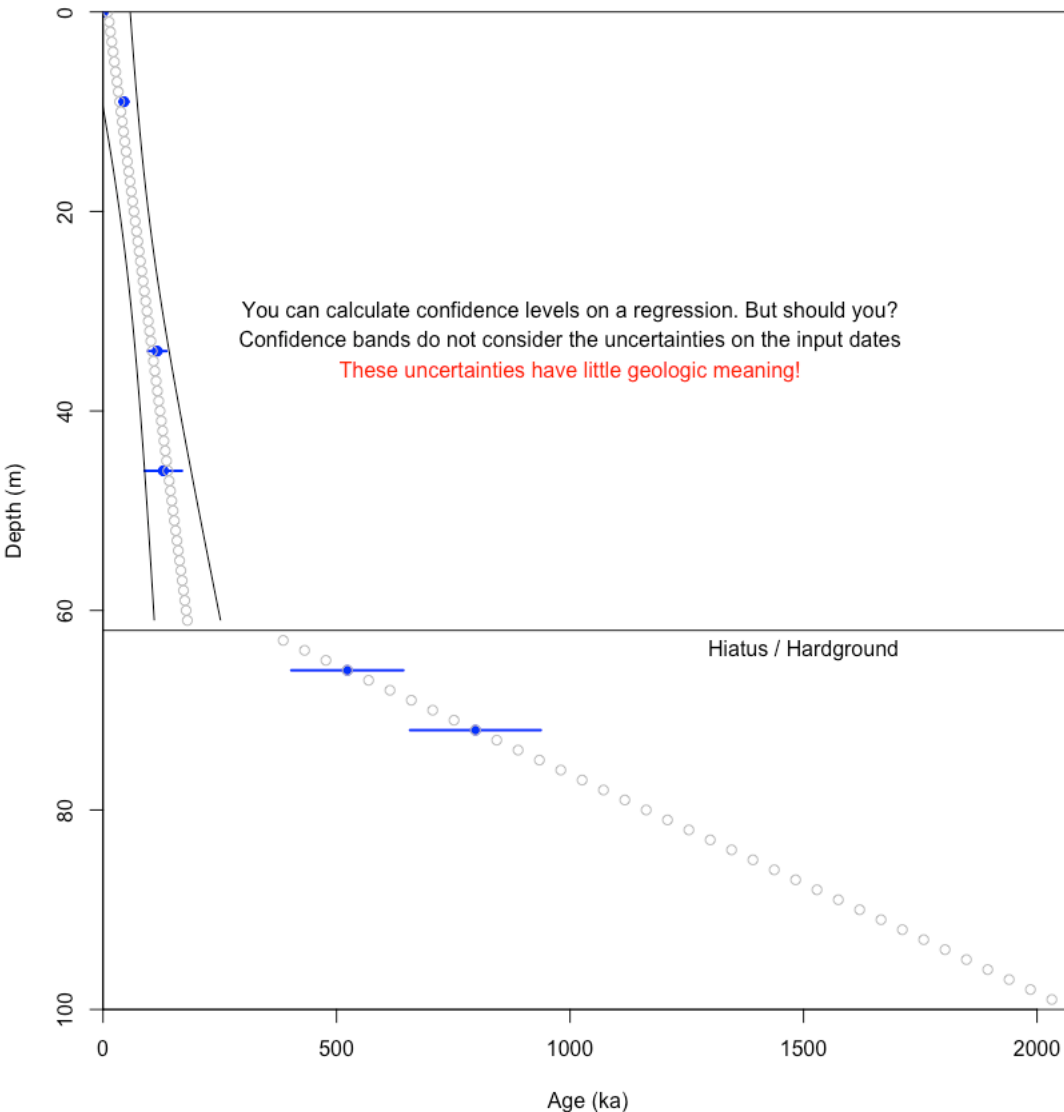
...but so does assuming a hiatus at 62 m.



**Linear interpolation
and extrapolation**

→ 62 m = hiatus

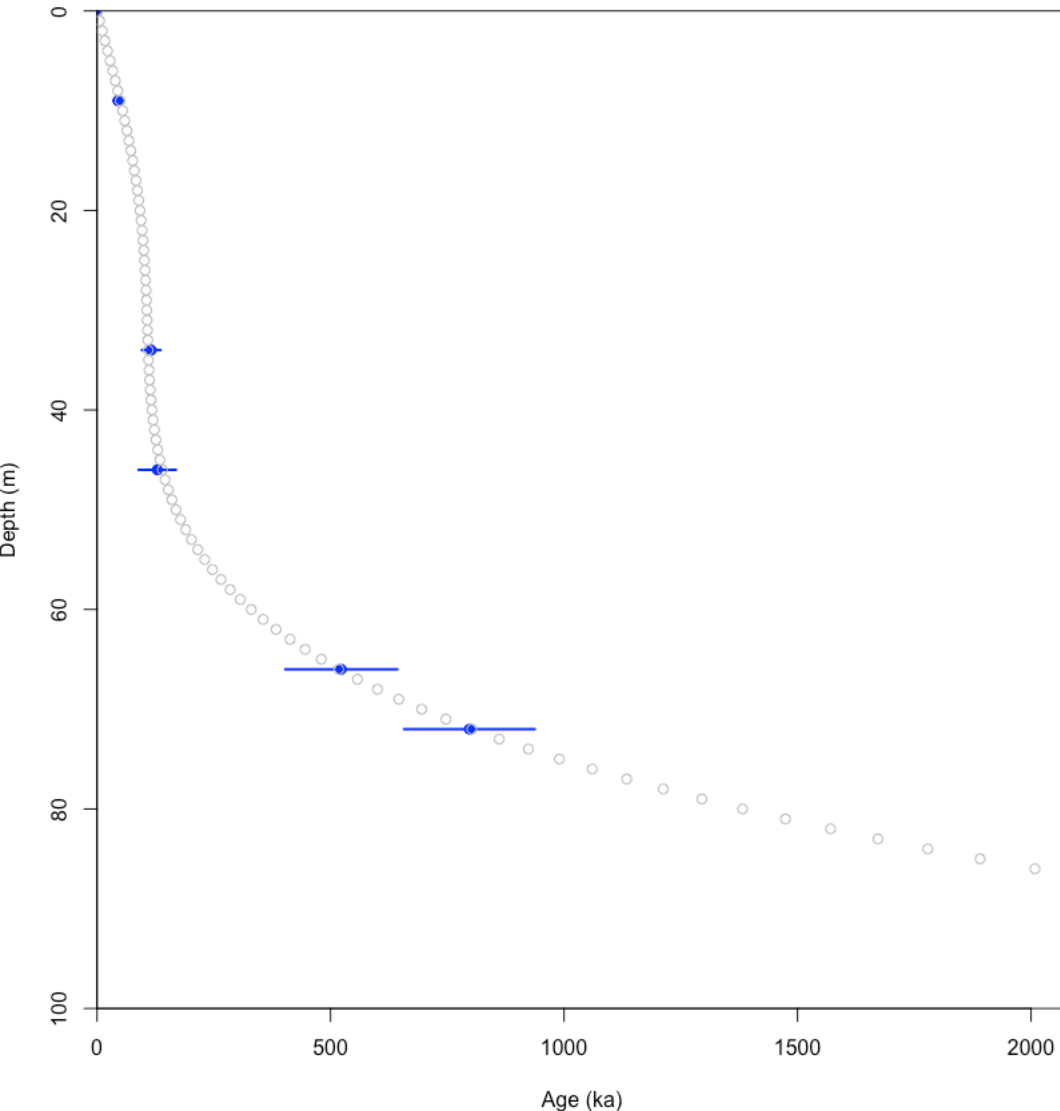
Linear regression puts somewhat less weight on individual dates, but the model is not forced through 0.



Linear regression
62 m = hiatus

**Uncertainties on
regression have little
geologic meaning**

Polynomial regression has narrow errors between ages but wide errors when extrapolating.



**Polynomial regression
(here 4th order)**

**62 m = hiatus
Fourth order**

**Crazy (low/high)
sedimentation rates
when extrapolating!**

**Law of superposition
not always respected**

Basic age-modelling have the advantage of simplicity but have some **drawbacks and limitations**.

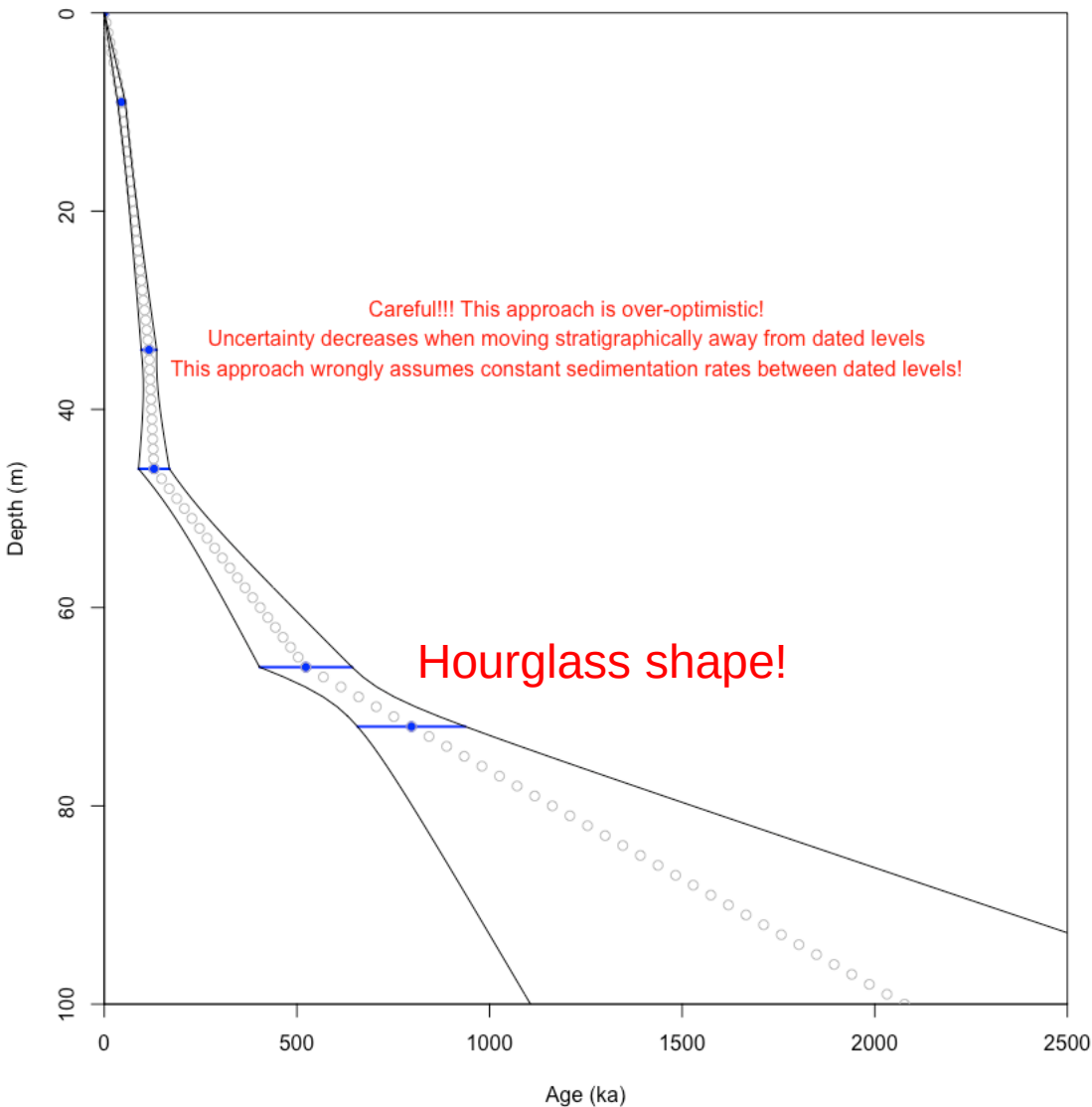
- One chooses the model that looks nicest... No transparent process.
 - Outlier or hiatus?
 - Crazy sedimentation rates?
 - Violation of the “Law of superposition”.
- Usually, not very well documented in literature.

MCMC

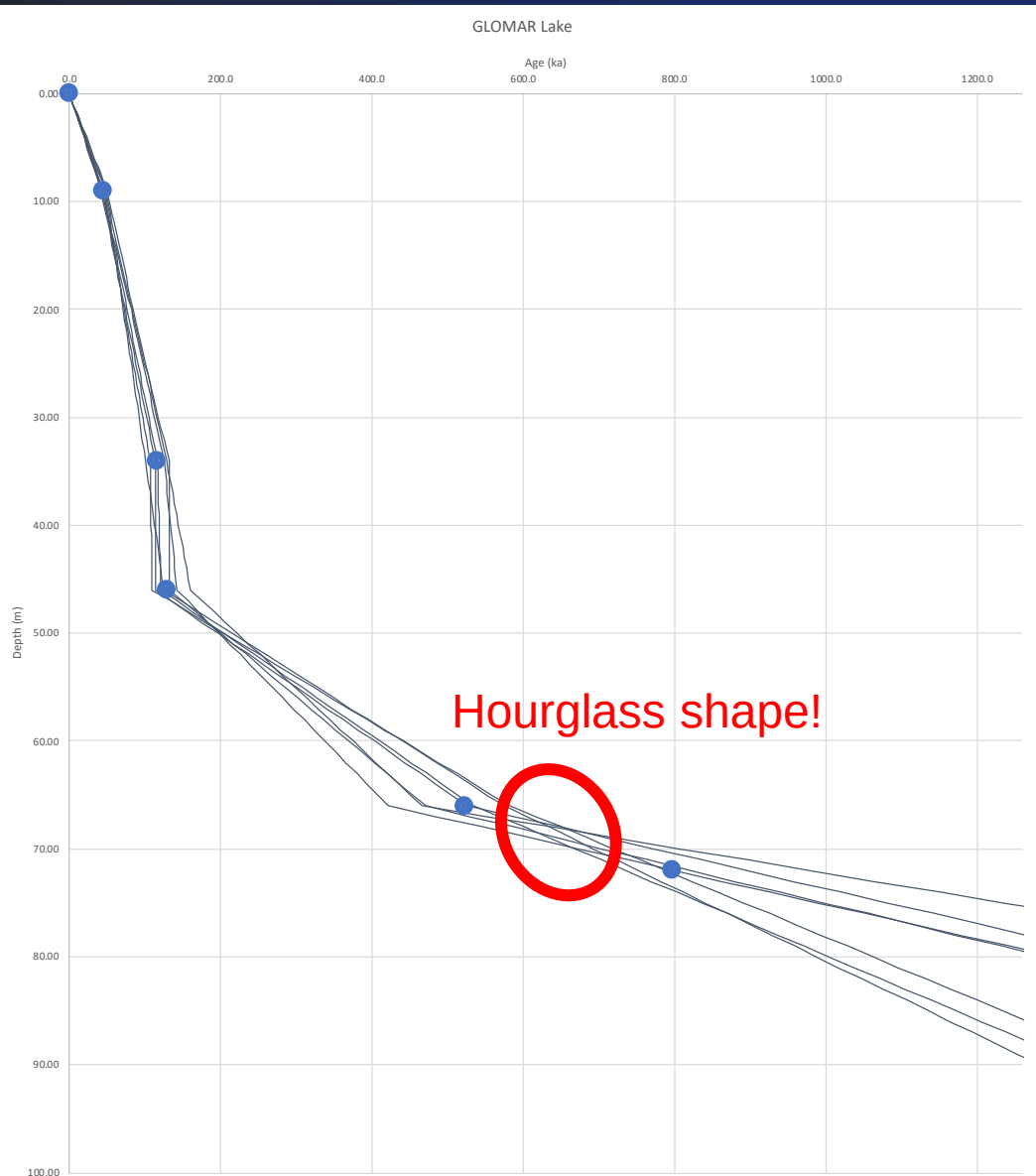
Monte Carlo simulation is a computational, probabilistic technique that predicts possible outcomes of uncertain events by running thousands of simulations using random sampling.

A Markov chain is a stochastic model describing a sequence of possible events where the probability of moving to the next state depends solely on the current state, not on previous history "memoryless".

Analytical error propagation... a dangerous pitfall!

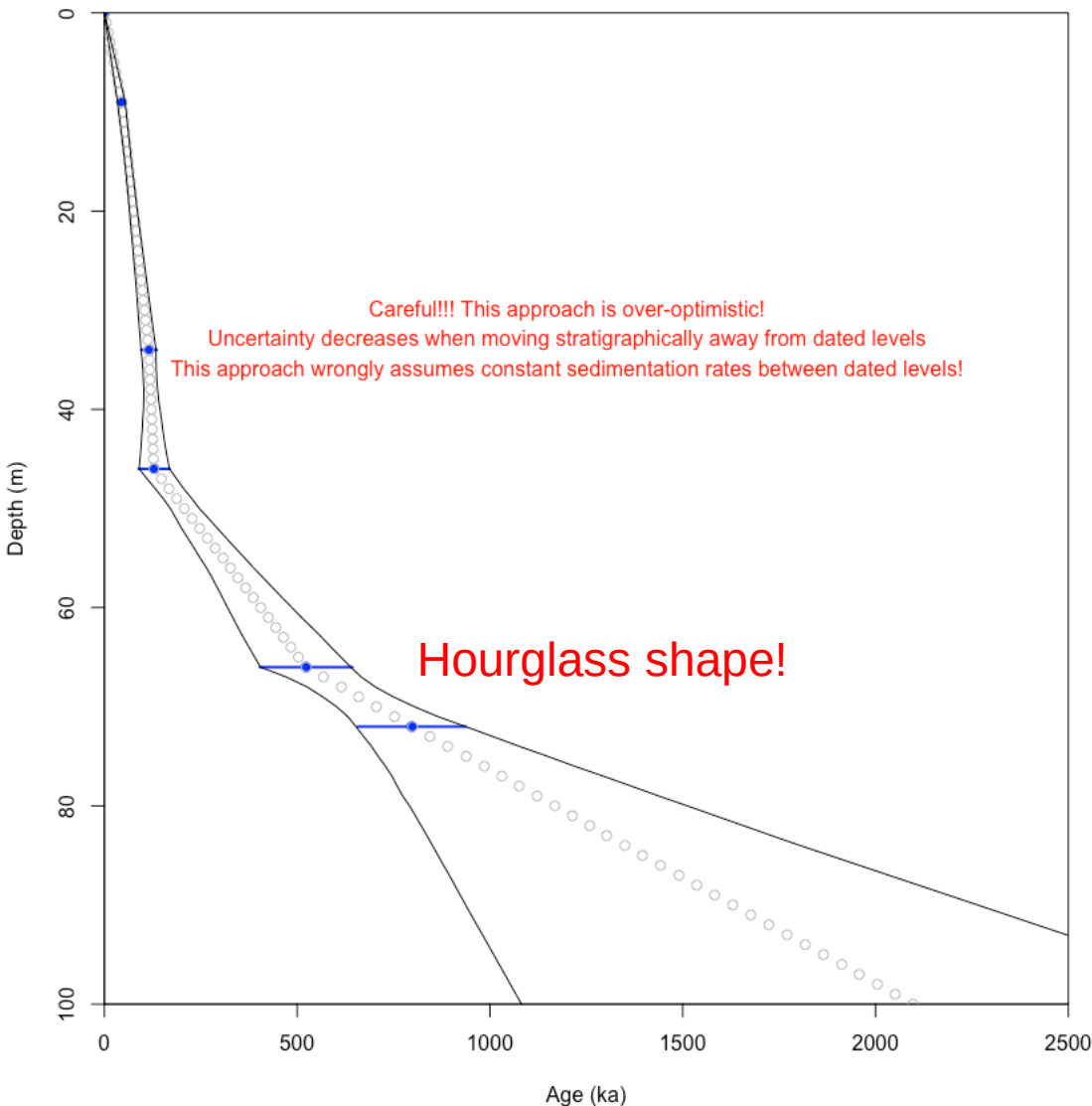


Monte Carlo approaches to basic age models also need careful use!



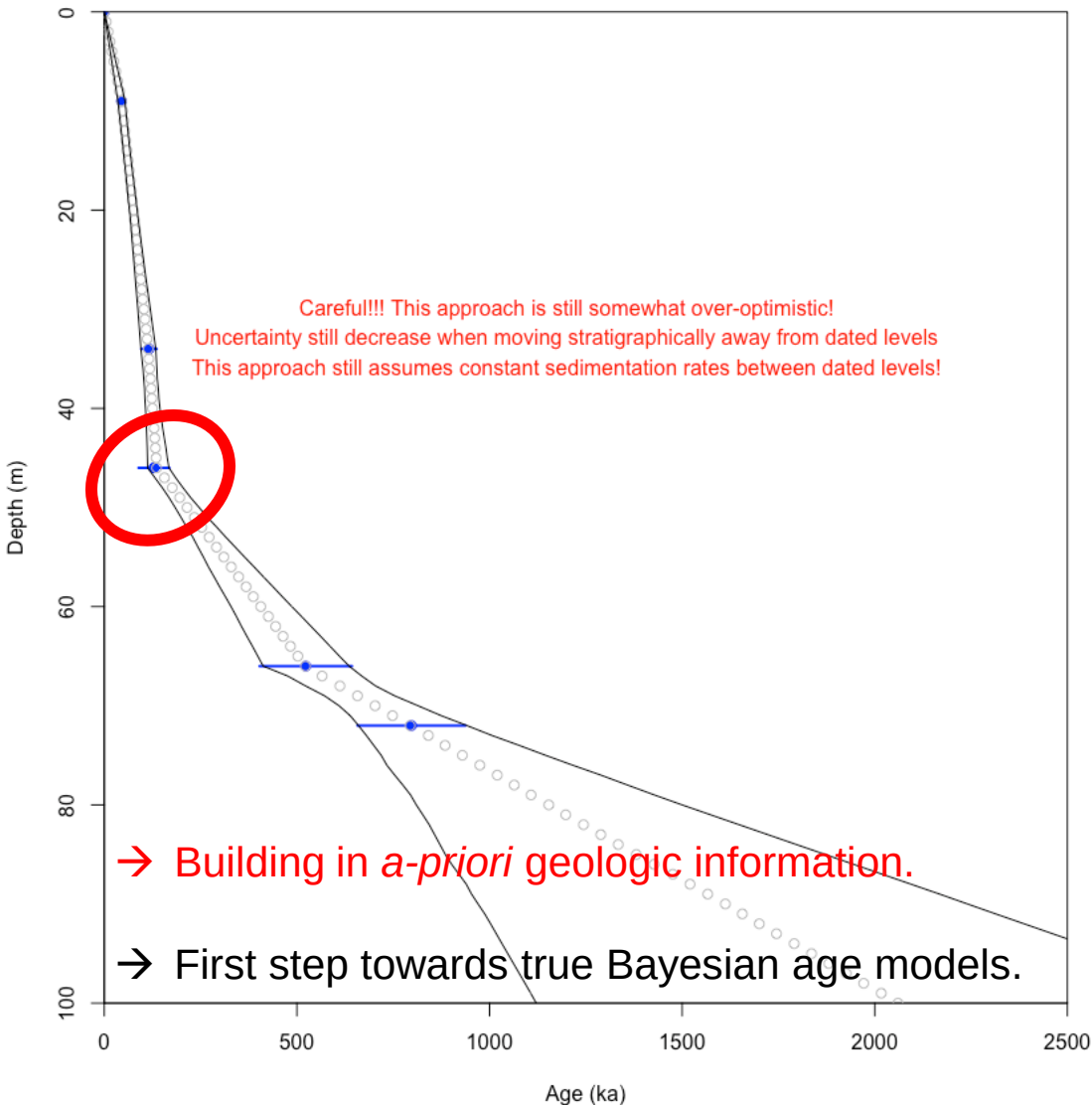
- i) At every dated level, an age is sampled from its probability distribution (here: normal distribution)
- ii) An age-model is drawn through these points (here: linear interpolation)
- iii) For every depth (here 1 m steps) the resulting age is stored.
- iv) Repeat many times (here: only 7 simulations shown, but you should do 1,000s)
- v) Calculate 2.5%, 50%, 97.5% percentiles for every depth.

Monte Carlo approaches to basic age models also need careful use!
The result is quasi-identical to the analytic uncertainty propagation.



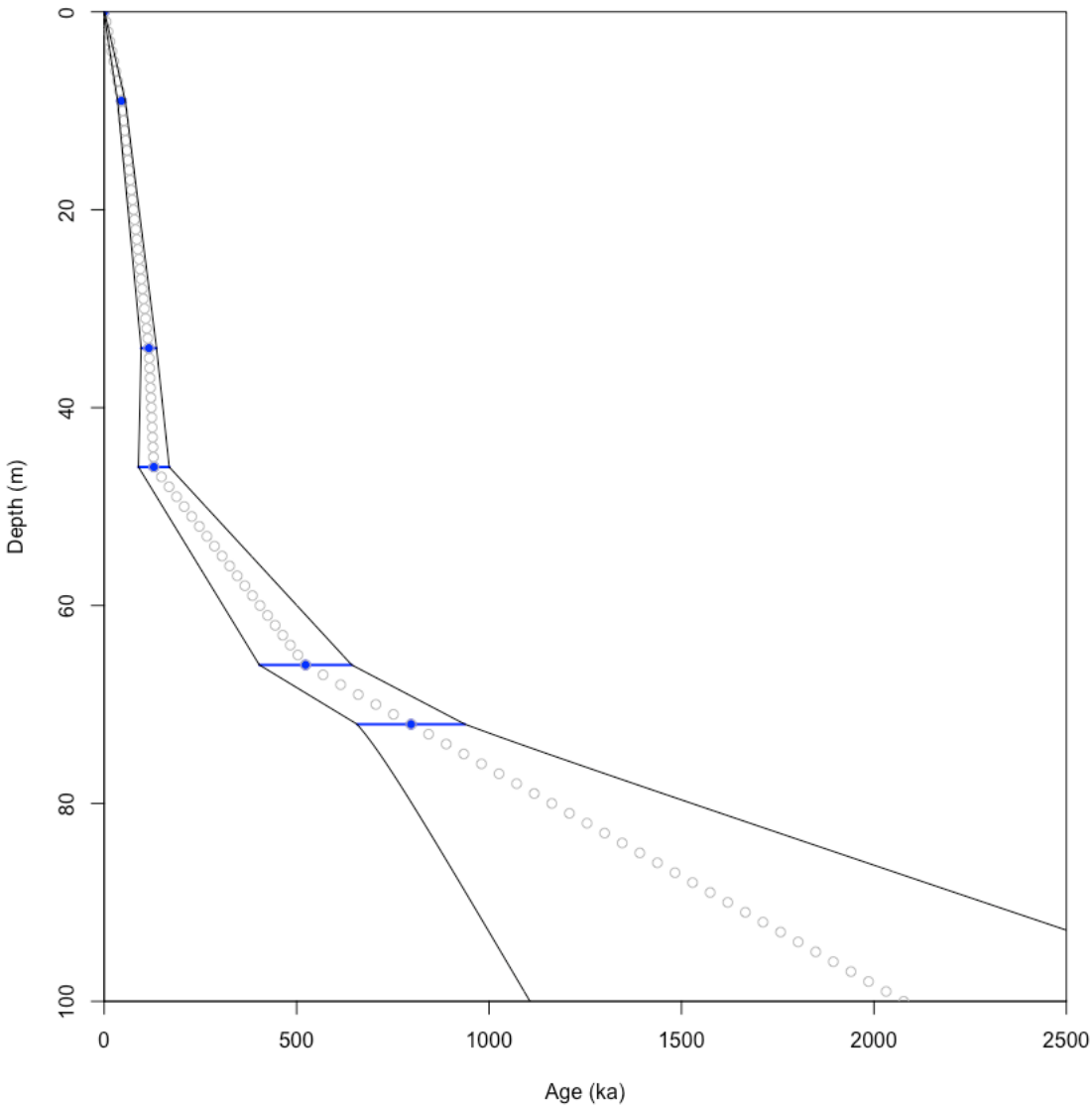
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Monte Carlo simulations with “age inversions” are thrown out. A first step to Bayesian statistics!



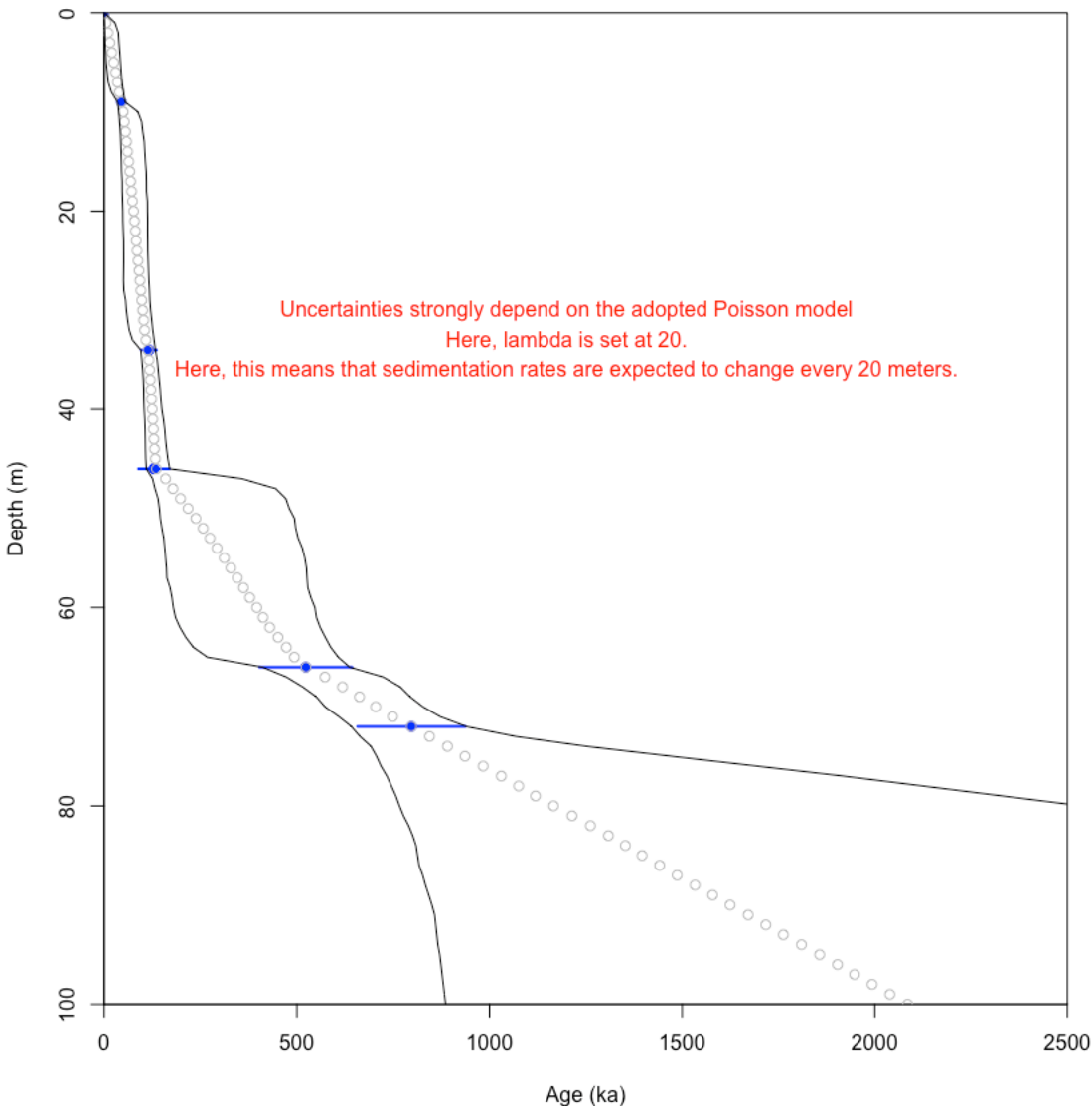
- i) At every dated level, an age is sampled from its probability distribution (here: normal distribution)
- ii) Verify whether sampled ages increase monotonically. Ignore sampled chronologies with age inversions.
- iii) An age- model is drawn through these points (here: linear interpolation)
- iv) For every depth (here 1 m steps) the resulting age is stored.
- v) Repeat many times (here: only 7 simulations shown, but you should do 1,000s)
- vi) Calculate 2.5%, 50%, 97.5% percentiles for every depth.

But..., you can also keep it simple by interpolating uncertainty levels



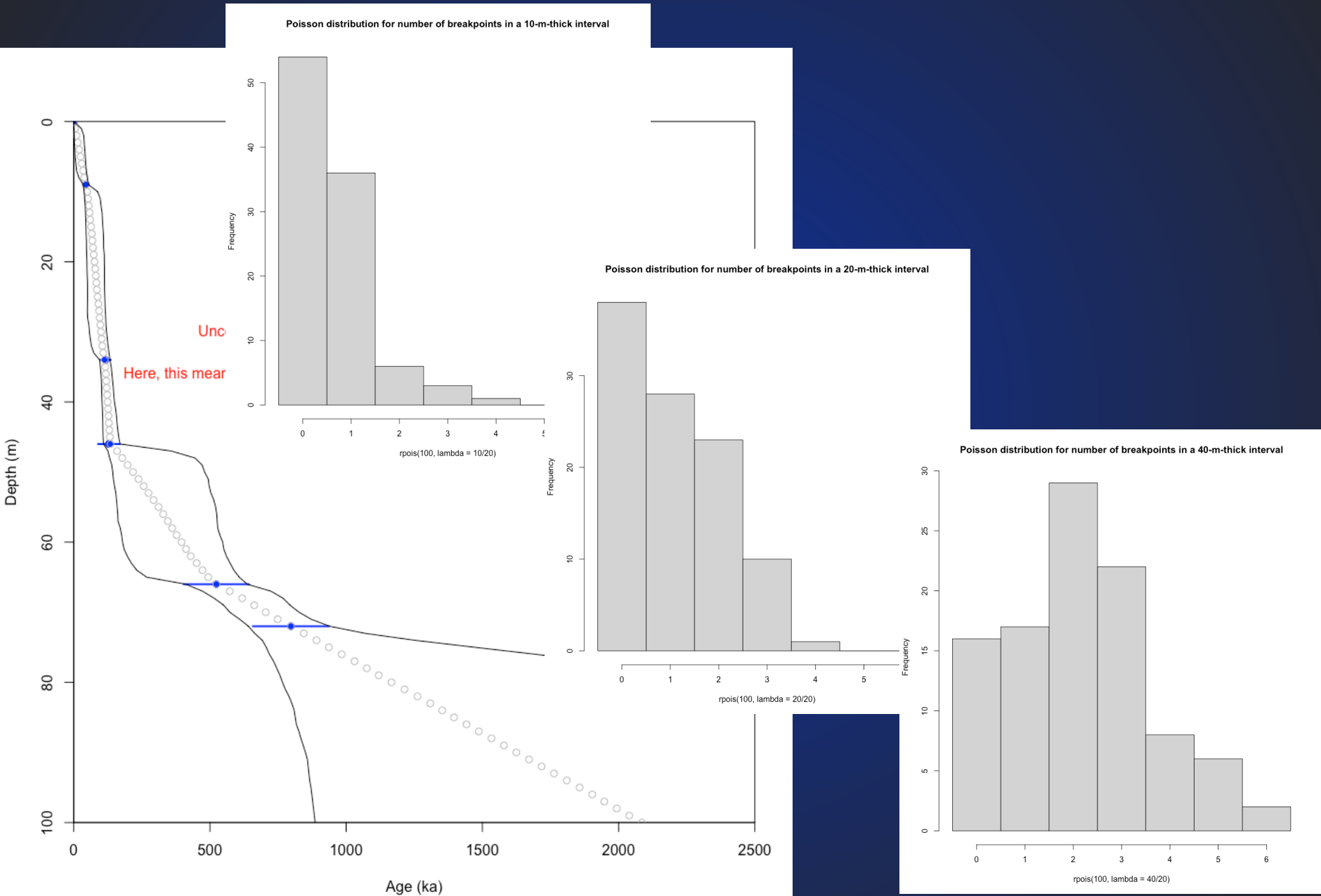
Tipping our toes in Bayesian approaches!

Adding sedimentation rate break-points in-between dated levels



- i) At every dated level, an age is sampled from its probability distribution
- ii) Throw-out age-inversions
- iii) Use a Poisson-distribution to add sedimentation break-points in-between dated levels. Sample a depth and age for those added levels (here: uniform distribution)
- iv) An age- model is drawn through these points (here: linear interpolation)
- v) For every depth (here 1 m steps) the resulting age is stored.
- vi) Repeat many times (here: only 7 simulations shown, but you should do 1,000s)
- vii) Calculate 2.5%, 50%, 97.5% percentiles for every depth.

Adding sedimentation rate break-points in-between dated levels



Summary classical age-depth models

- Interpolation and extrapolation
- Linear regression
- Polynomial regression
- Different approaches to uncertainty

All age-depth models are based on certain assumptions. The key is to be aware of all assumptions that go into yours!

Software: selection

Classic methods: (regression, interpolation, spline):
Excel, (MATLAB), **R**

Bayesian:

Bchron - Haslett and Parnell (2008), Parnell 2016

Bacon - Blaauw & Christen, 2018

ADMin - Zeeden et al. 2018

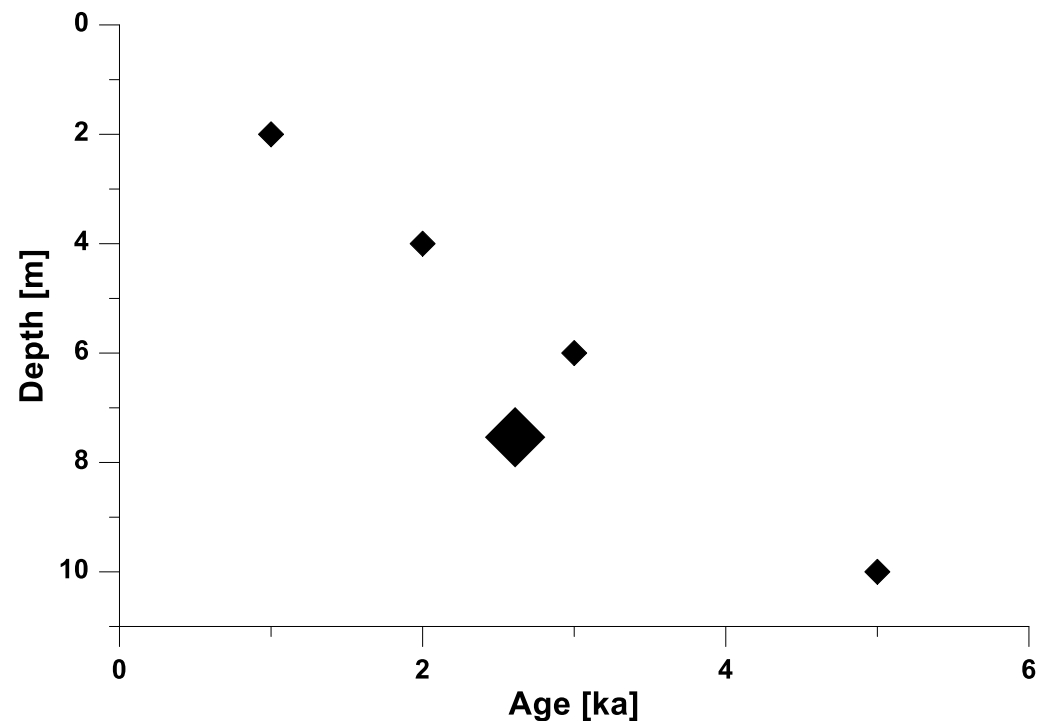
ArchaeoChron (Philippe & Vibet 2017), dependent on JAGS

OxCal (e.g. Bronk Ramsey 2009): online, distribution

Further exits

Bayesian Modelling:

Enforcing stratigraphic consistency (no age inversions)



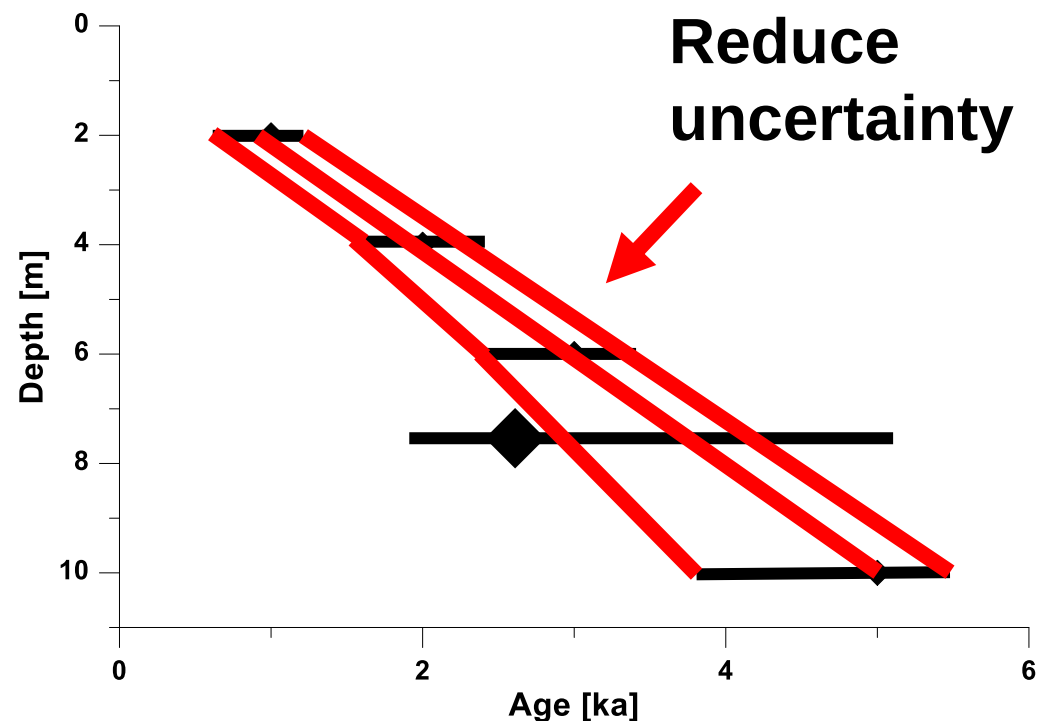
Bayesian Modelling:

Enforcing stratigraphic consistency (no age inversions)

sample ages
consistent?

Yes $\rightarrow$ keep

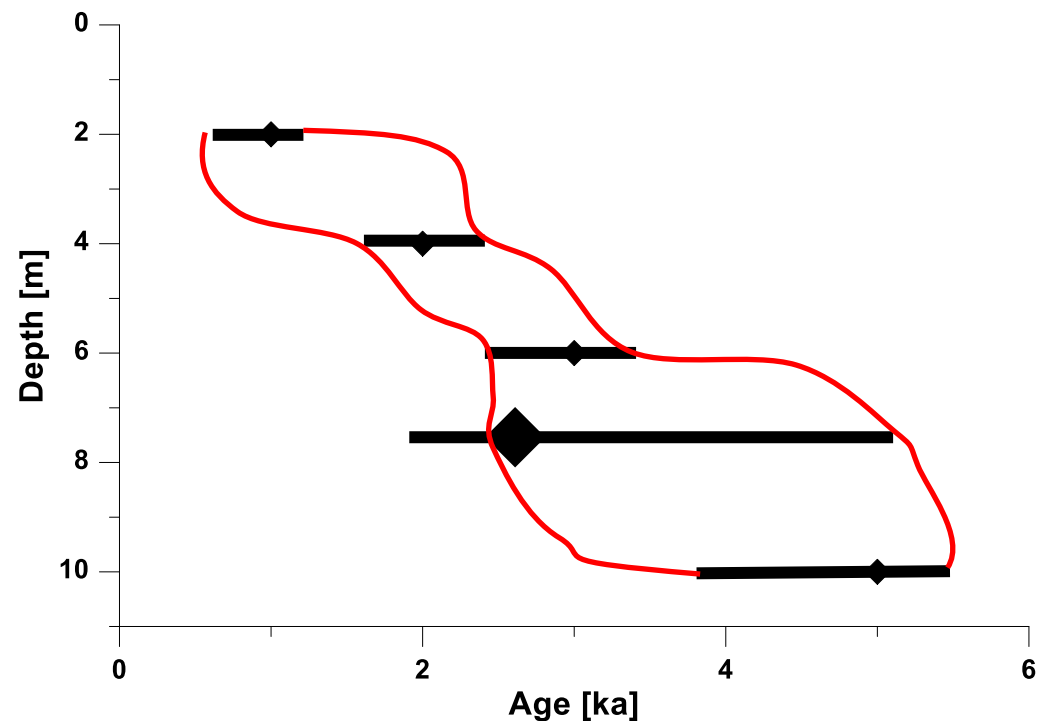
No $\rightarrow$ bin



Bayesian Modelling:

Enforcing stratigraphic consistency (no age inversions)

model SR
consistent?
Yes $\rightarrow$ keep
No $\rightarrow$ bin



Bayesian Modelling:
Avoiding long durations/keep
modelled age ranges realistic

Avoiding artefacts: dataset length

Avoiding long durations/keep modelled ages realistic: Please check your results

- add durations
- weight inverse to length
- add 'boundary ages'

2 model types: Assume Sed.rates, or not

Do we derive these, or use these as input?

- SR assumptions make model results more precise
- Is this more accurate? Does this reflect reality?

What happens between ages?

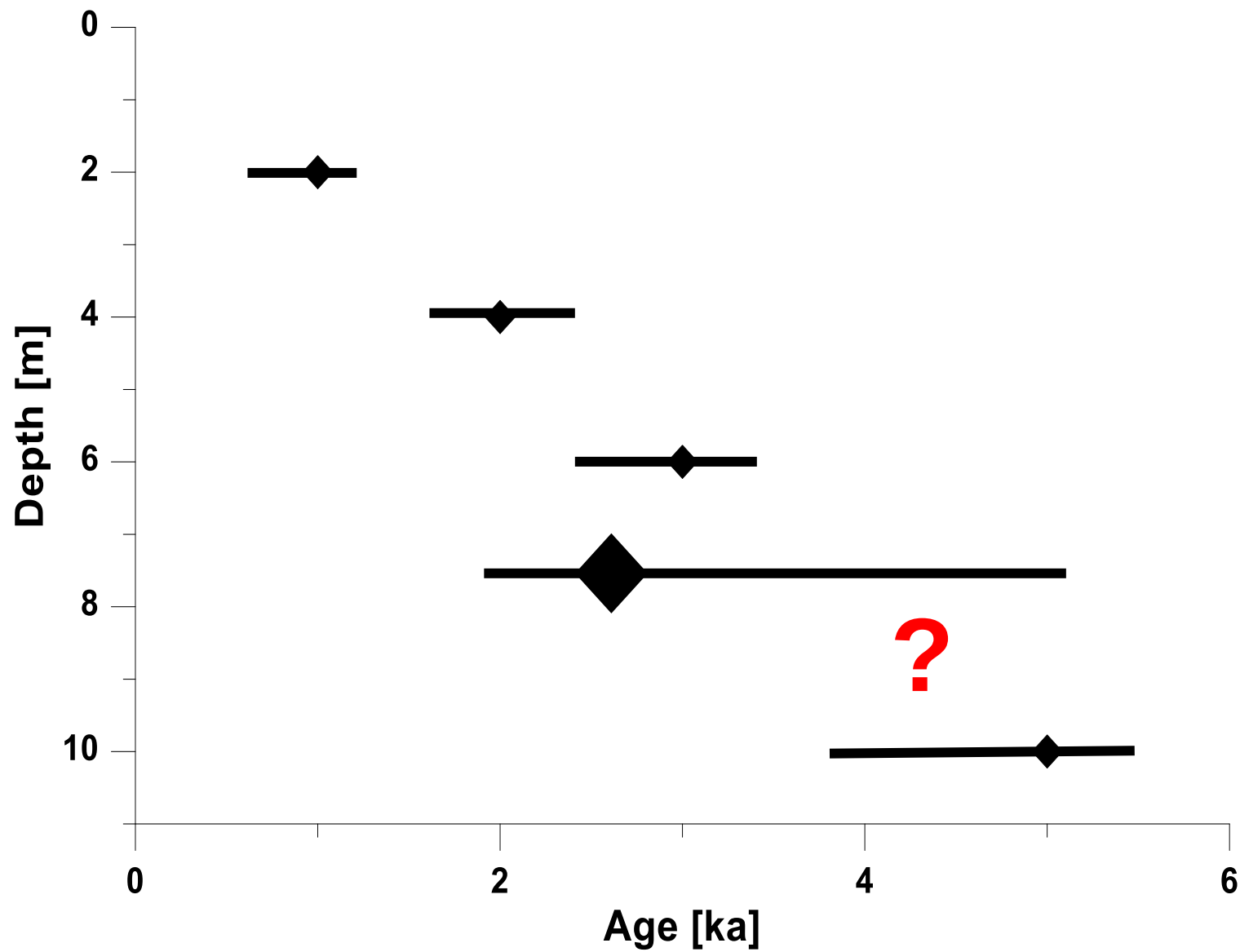
Reality & Models

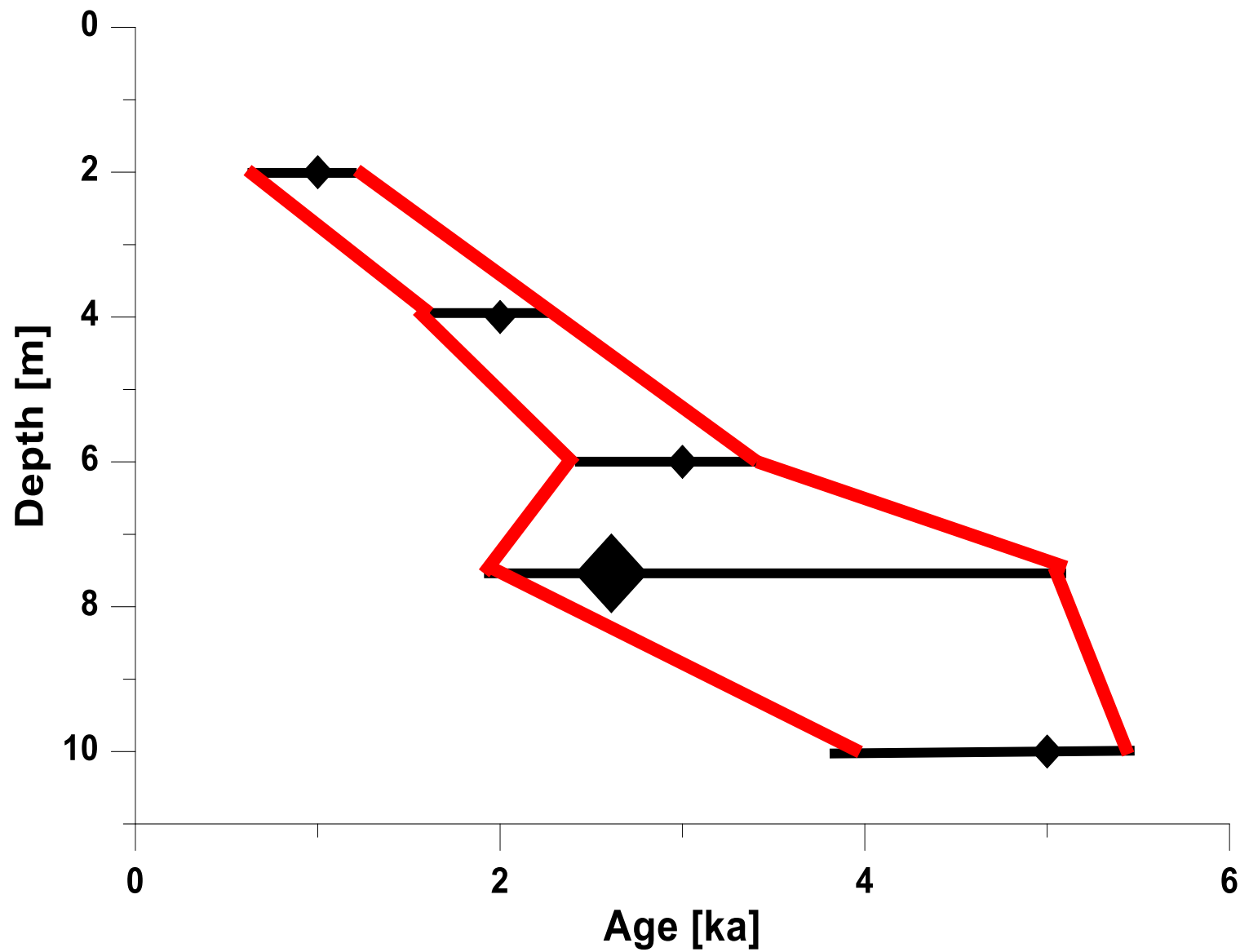
- In Reality we don't know
exceptions: durations in between: e.g. warves
- Different models exist with (very) different modelling outcome (more later)
- Here: 3 different age-depth models:
Bchron, (Bacon), ADMin

What happens between ages?

Reality & Models

- In Reality we don't know, simple solutions are:
- Not make statements on non-dated levels
- Interpolation of uncertainty
- 'at least' interpolation





What happens between ages?

Reality & Models

- In Reality we don't know
exceptions: durations in between: e.g. warves
- Different models exist with (very) different modelling outcome (more later)
- Here: 3 different age-depth models:
Bchron, (Bacon), ADMin

What is modelled between ages?

- Specific SR models/model outcome
- SR accumulation models by semi-infinite distributions/processes $[0:\infty]$
- e.g. Gamma
- Dependent on:
Number of intervals
Distribution parameters

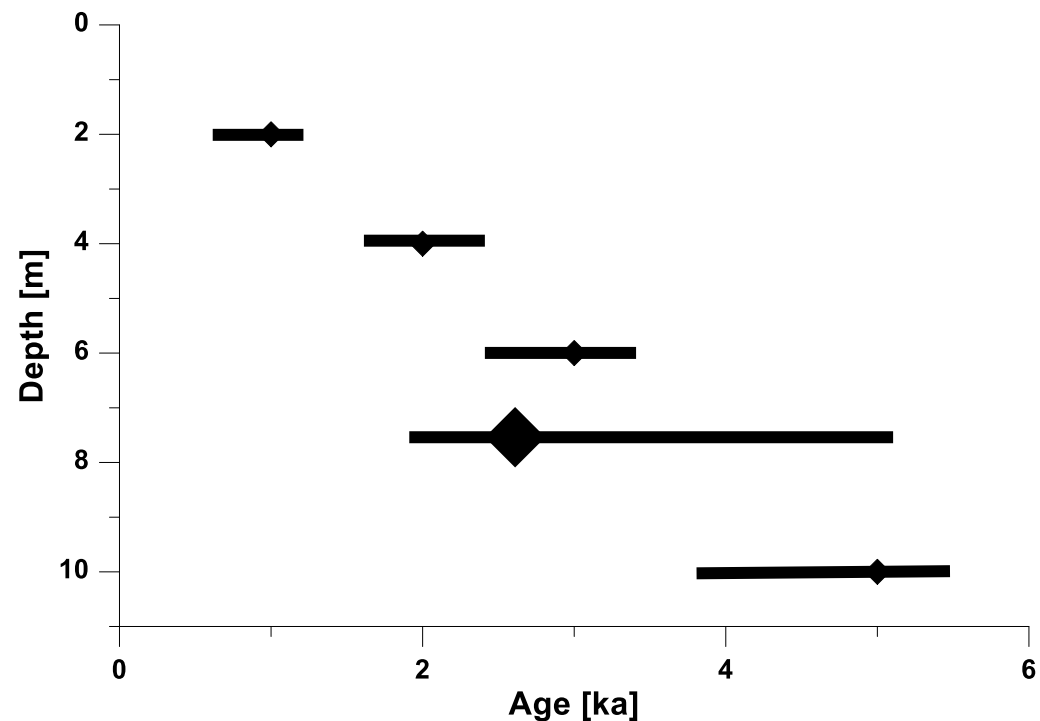
Uncertainty: relevant!

Enforcing stratigraphic consistency (no age inversions)

sample ages
consistent?

Yes → keep

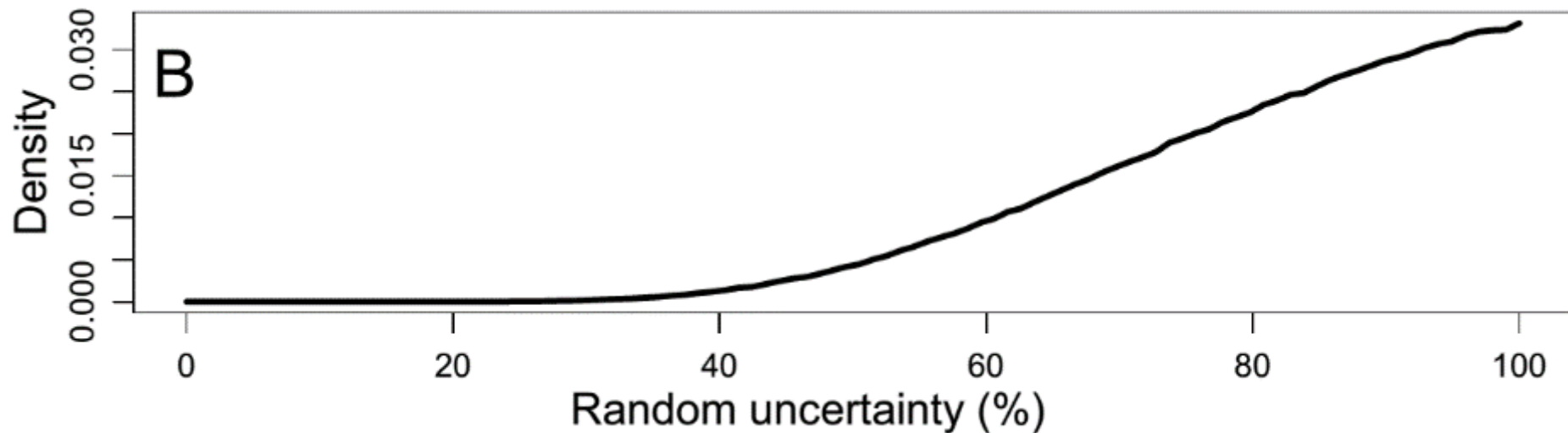
No → bin



Separation systematic/random

- Often challenging

Use only random uncertainty, recombine with systematic part
Implemented in ADMin



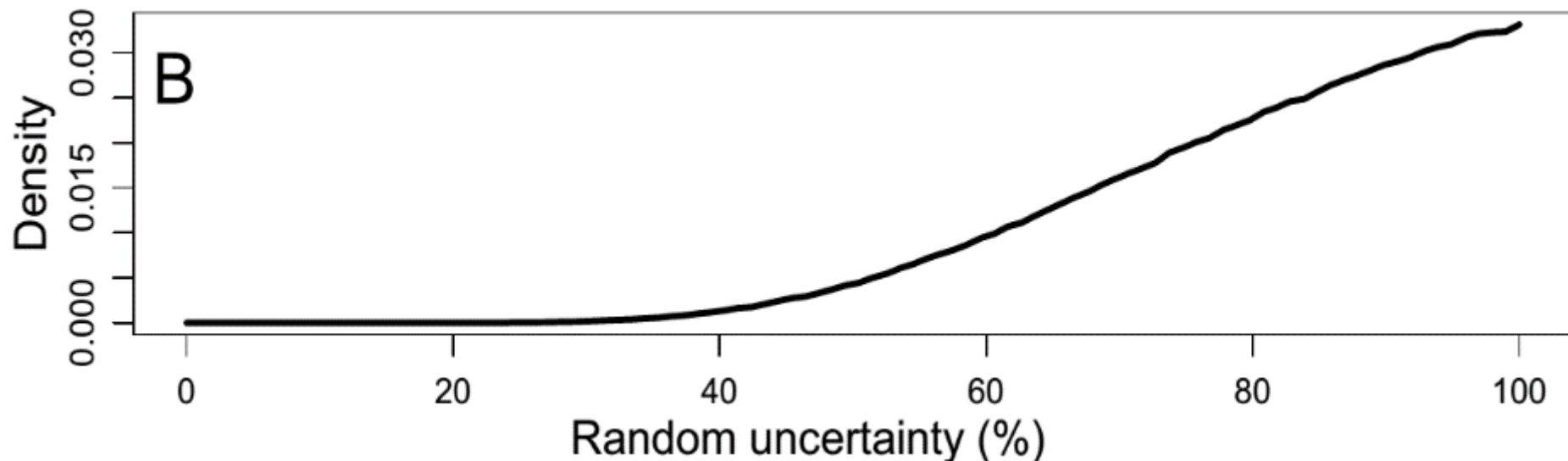
Separation systematic/random information depends on dating technique

Uncertainty: Random

→ age-depth models can improve precision

Uncertainty: Systematic

→ age-depth models can NOT improve precision



Summary:

- It can be complicated in detail
- Document well what you do
- Think about your results: do these make sense?
- Is duration longer than expected → likely model artefact

Going Bayesian - artificial experiment:

Depth: 0-100 cm (,core'/'outcrop')

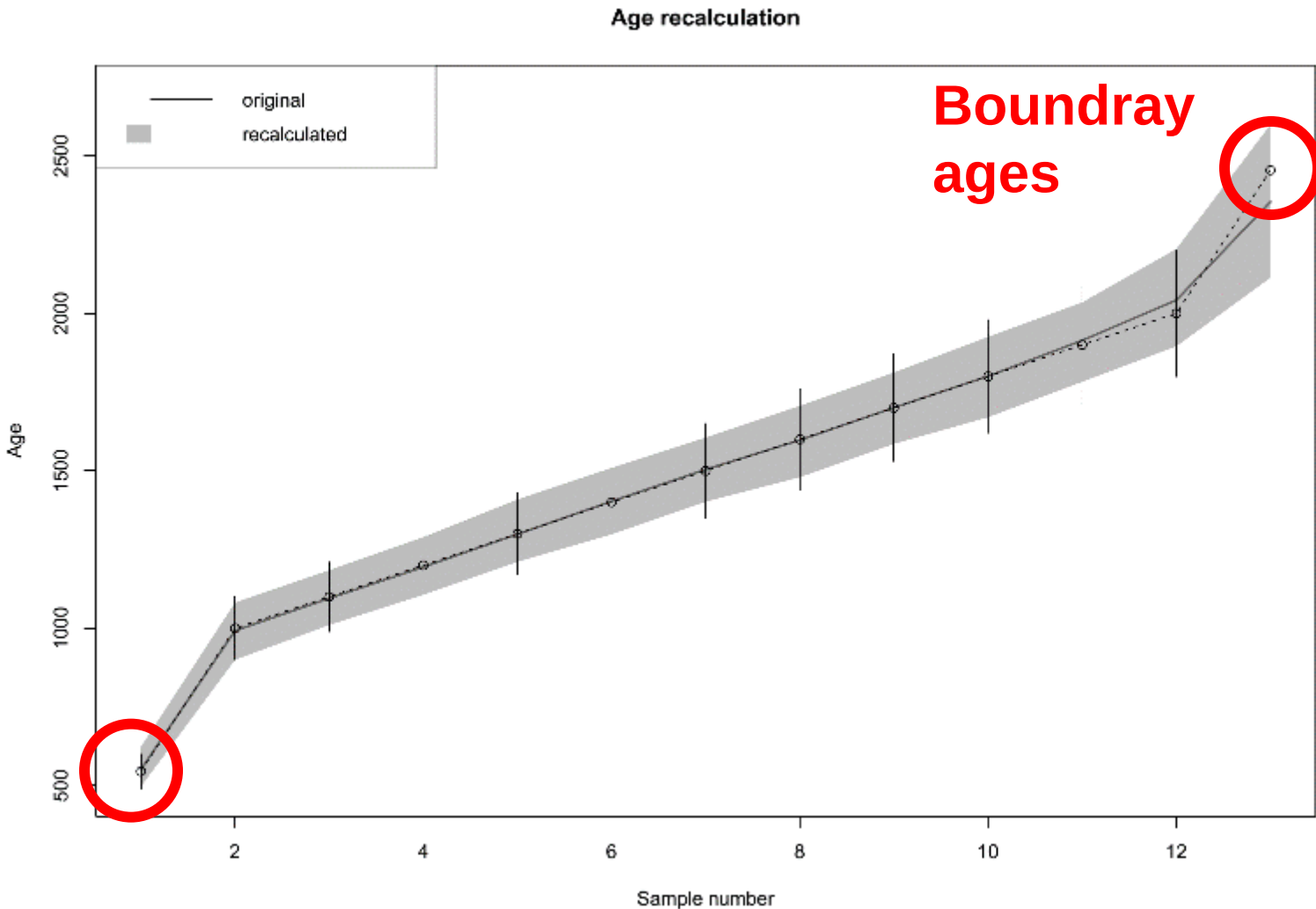
Age: 1000-2000 a (no age 0)

Uncertainty: set to 10% of age

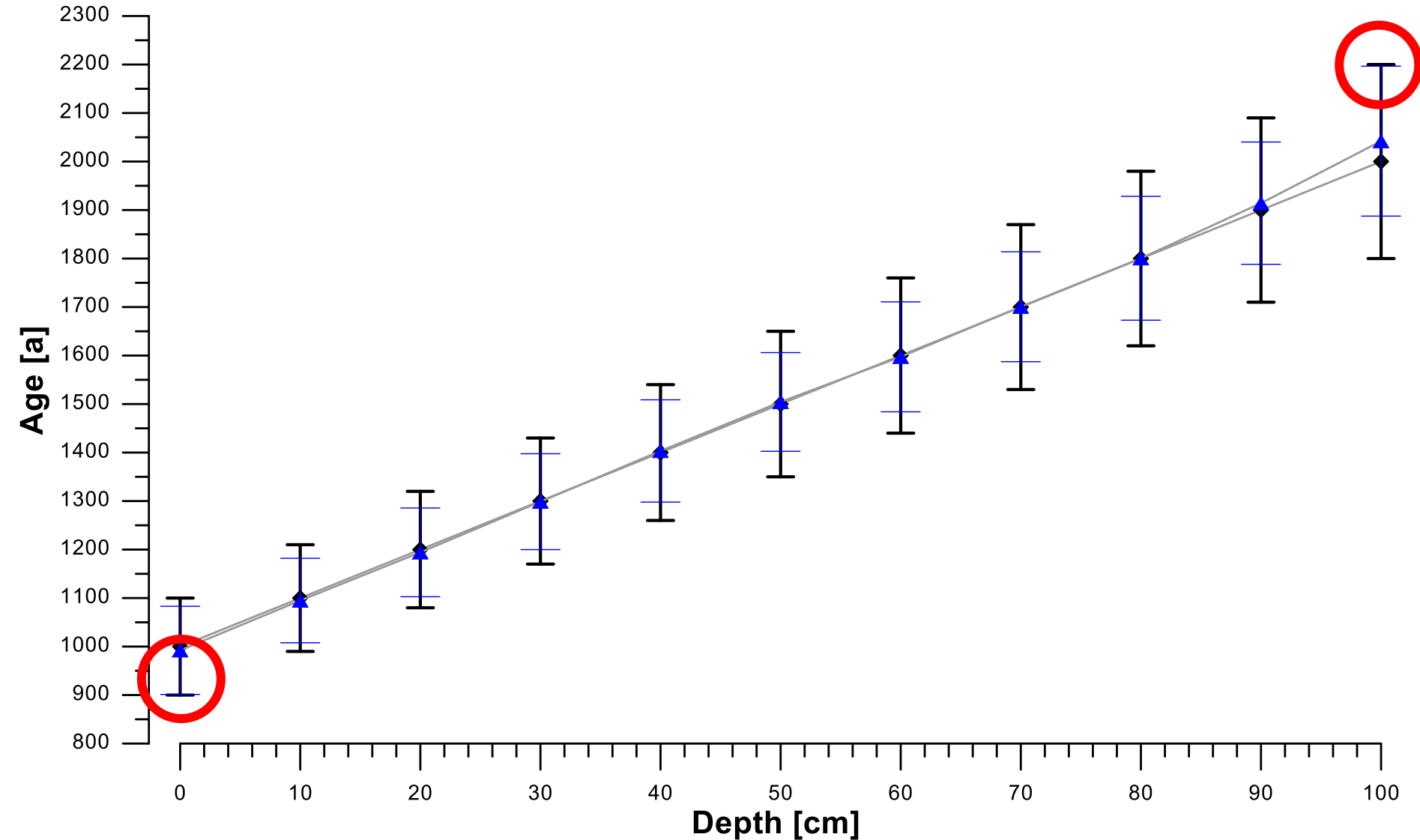
Every 10 cm/ 100a one sample/age

Given: raw model output

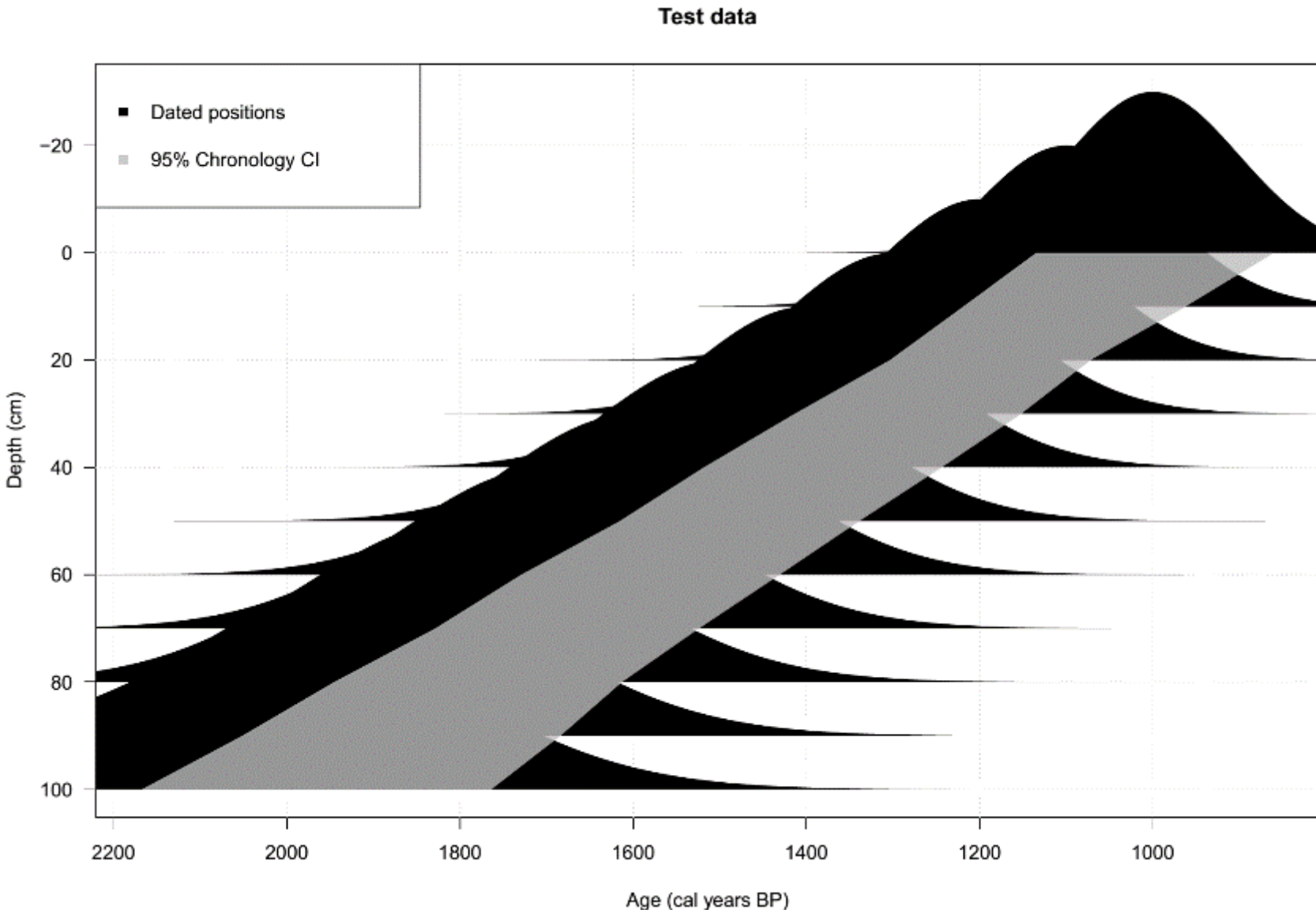
#1: ADMin: No SR model linear interpolation of quantiles



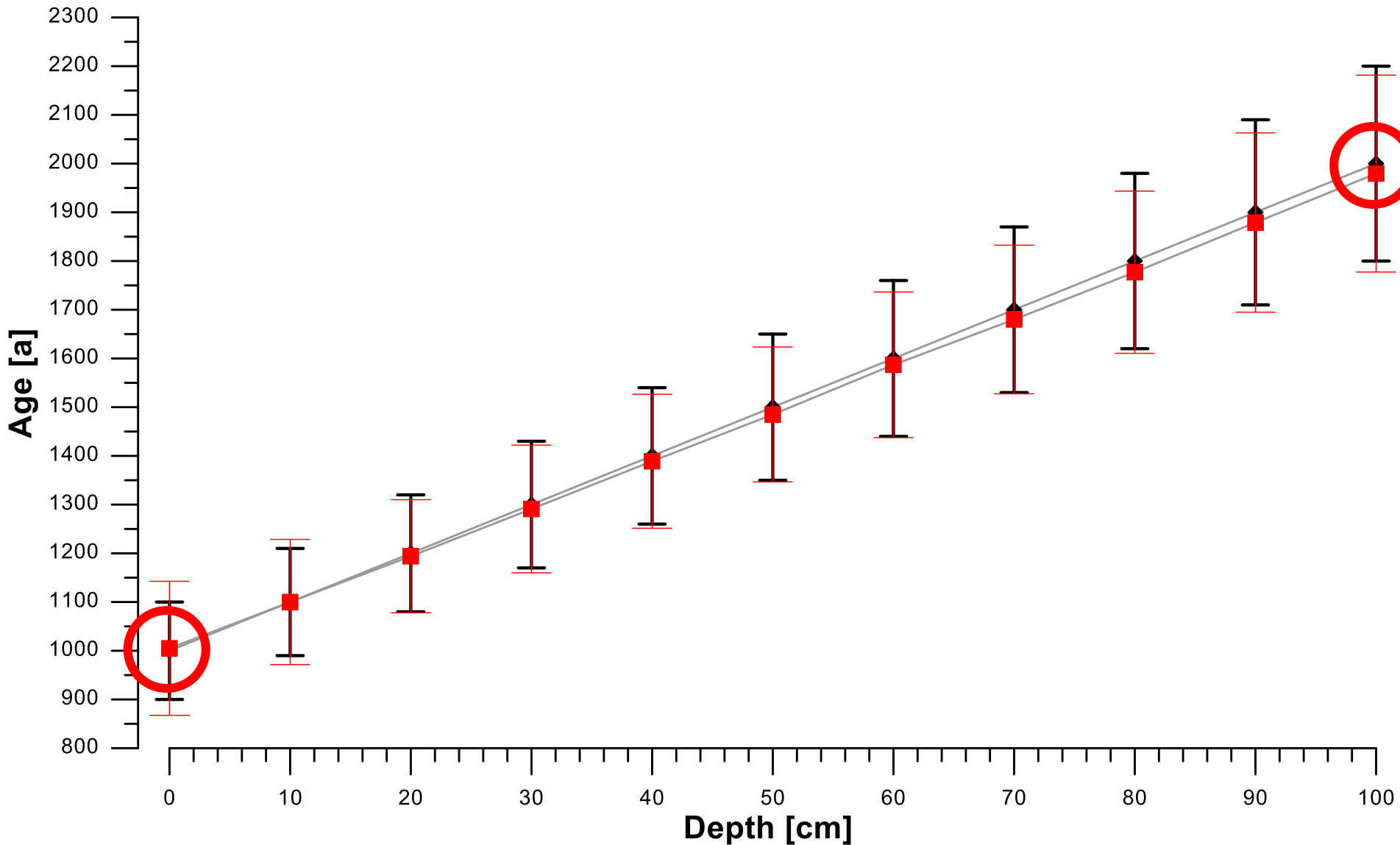
Test ages (black) vs. ADMin results (blue)



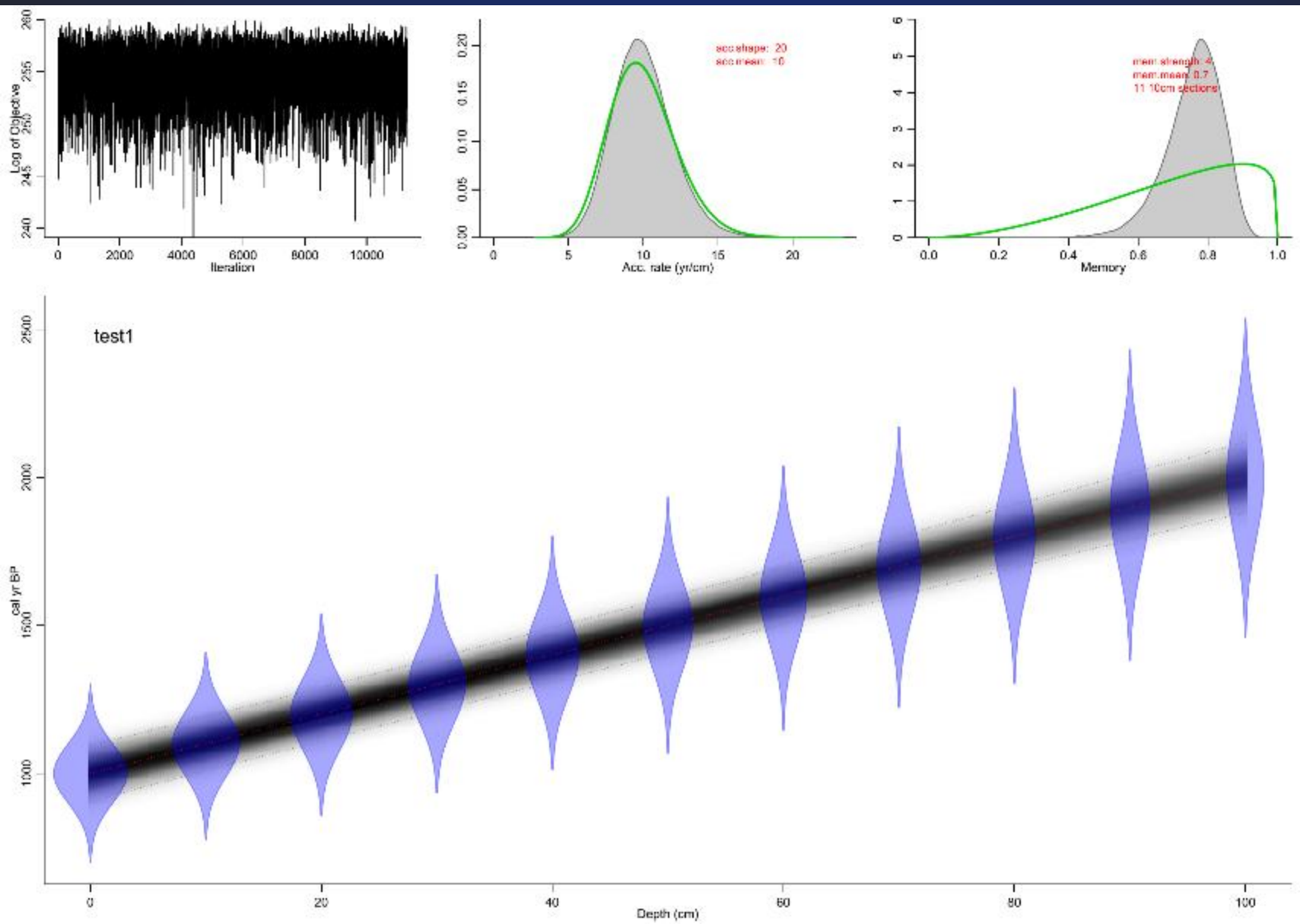
#1: Bchron: assumes linear SR



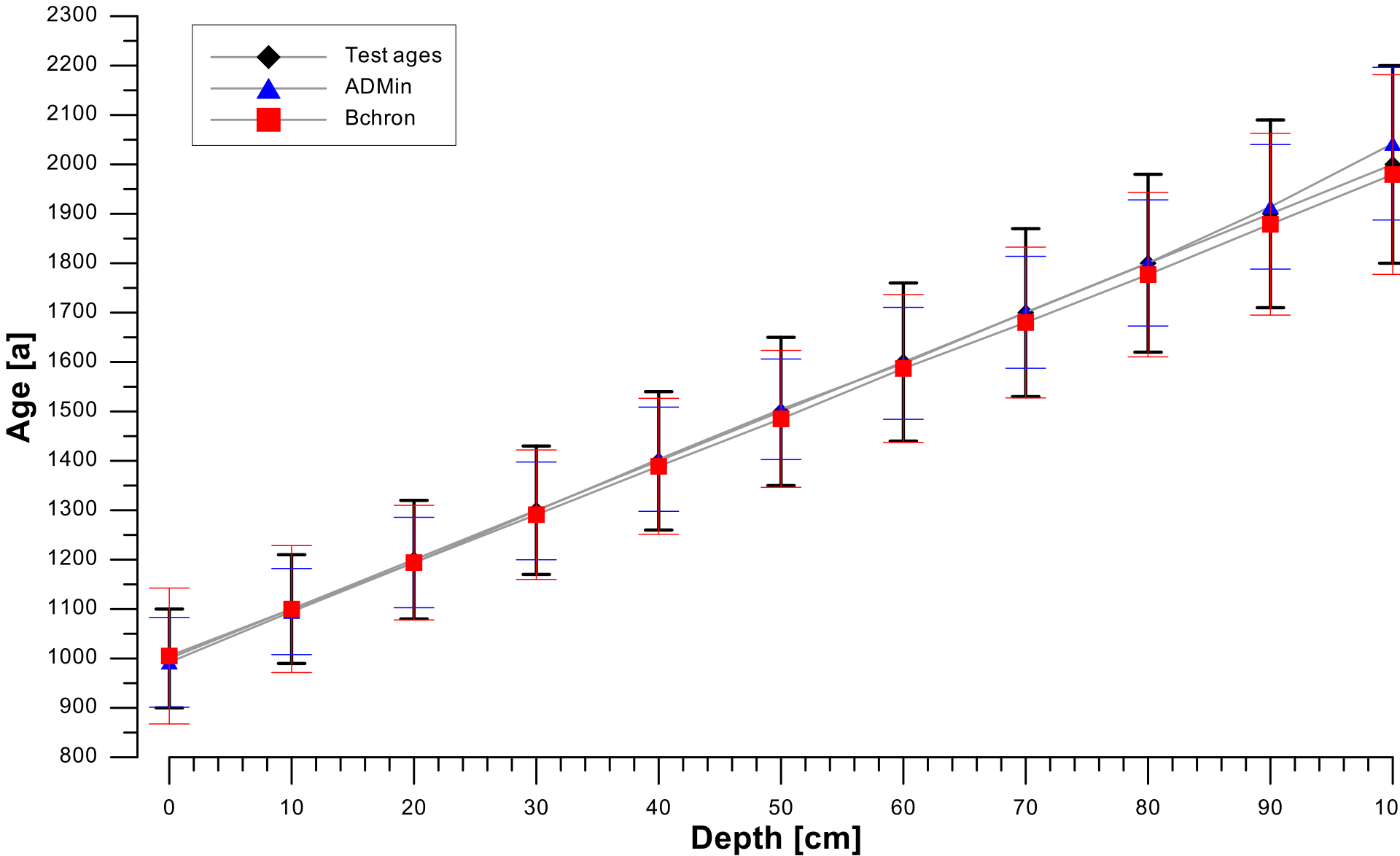
Test ages (black) vs. Bchron results (red)



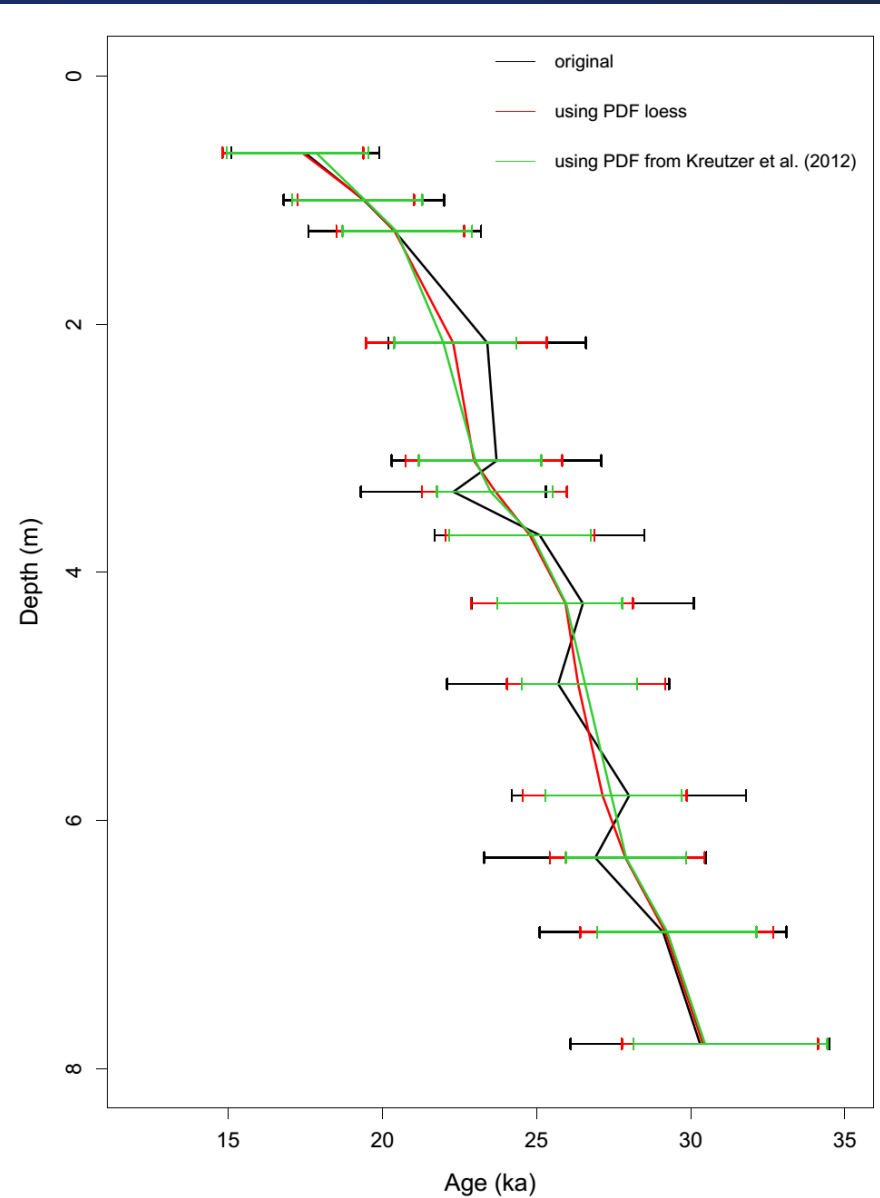
#1: Bacon: assumes mean SR



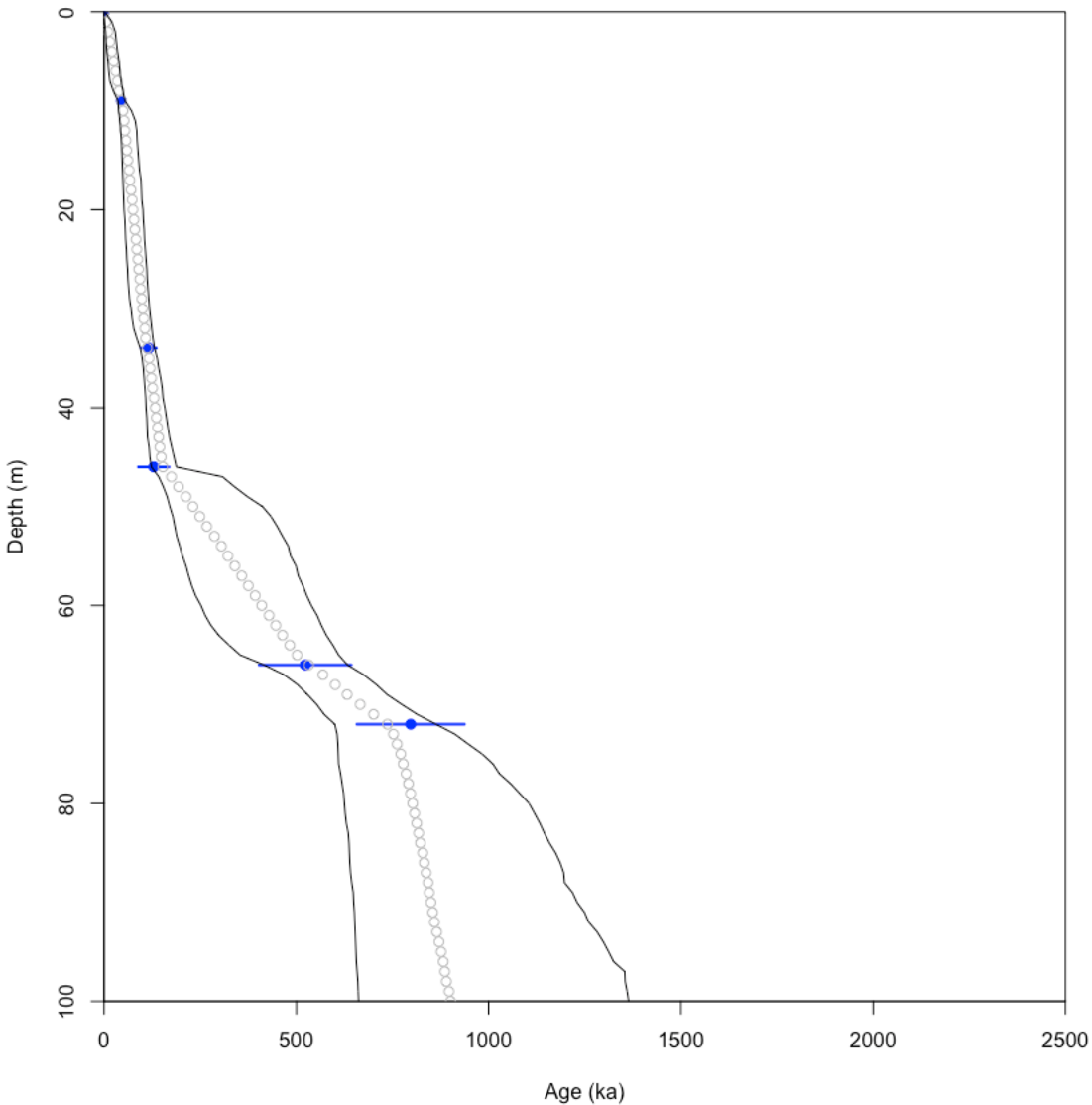
Comparison of methods



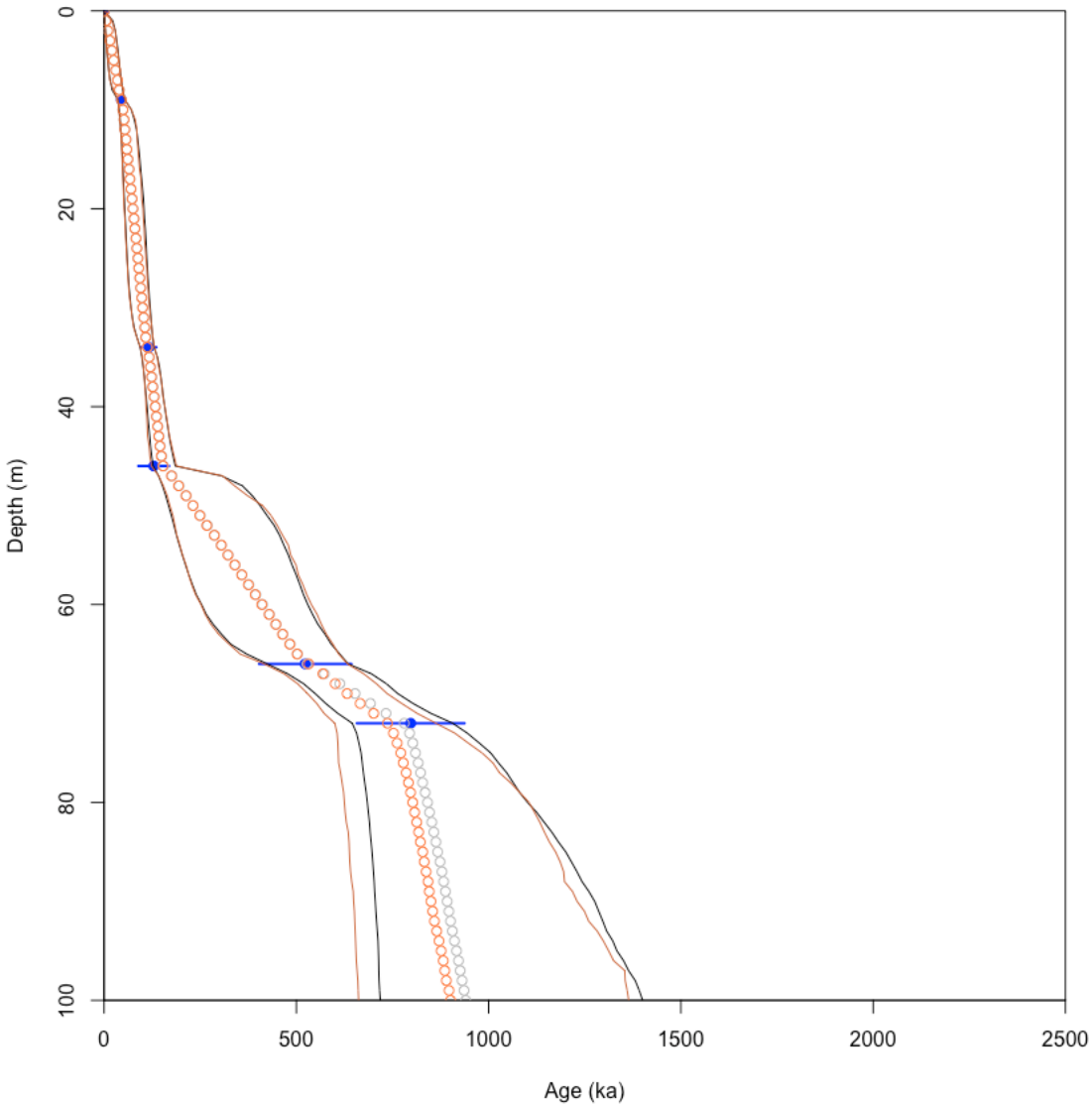
Example: OSL dating of Ostrau/Germany



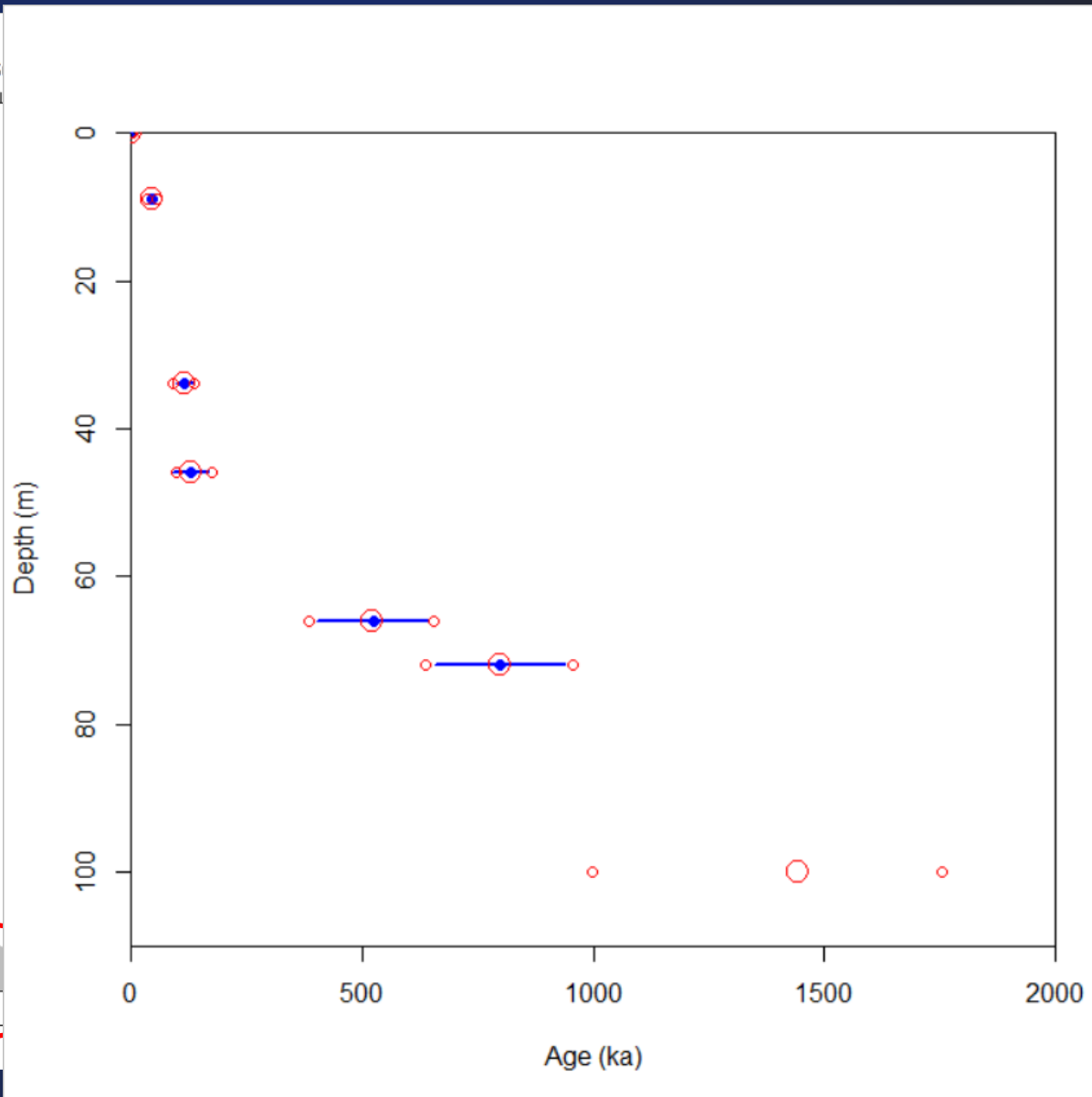
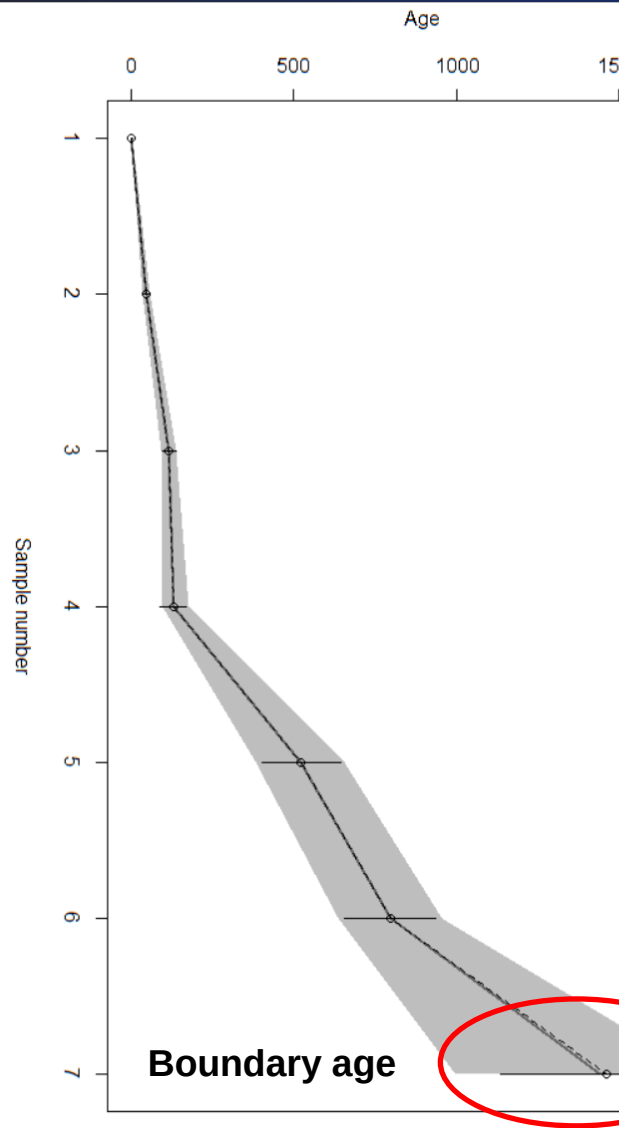
Bchron applied to GLOMAR Lake



modifiedBchron for non- ^{14}C dates



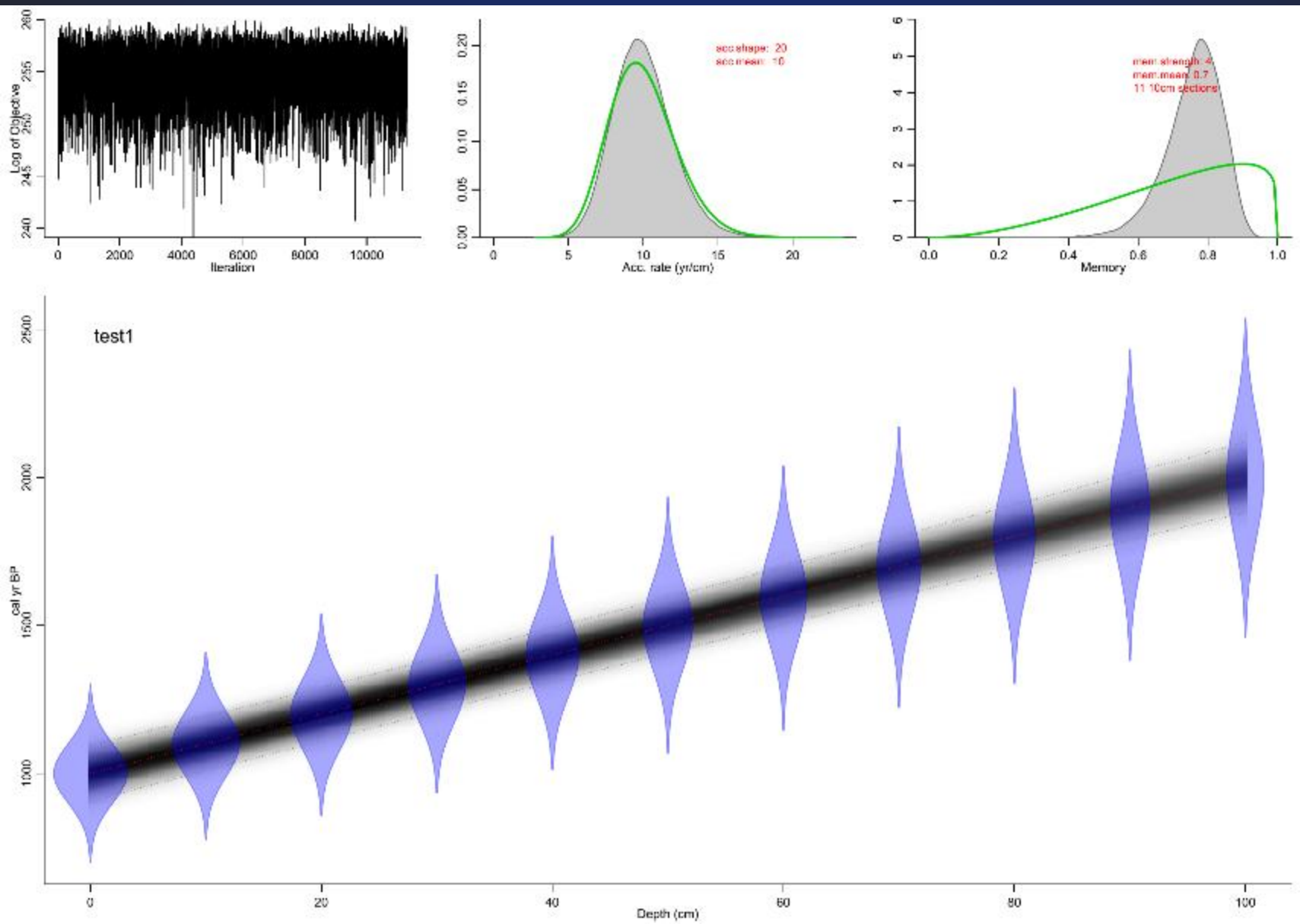
ADMin applied to GLOMAR Lake



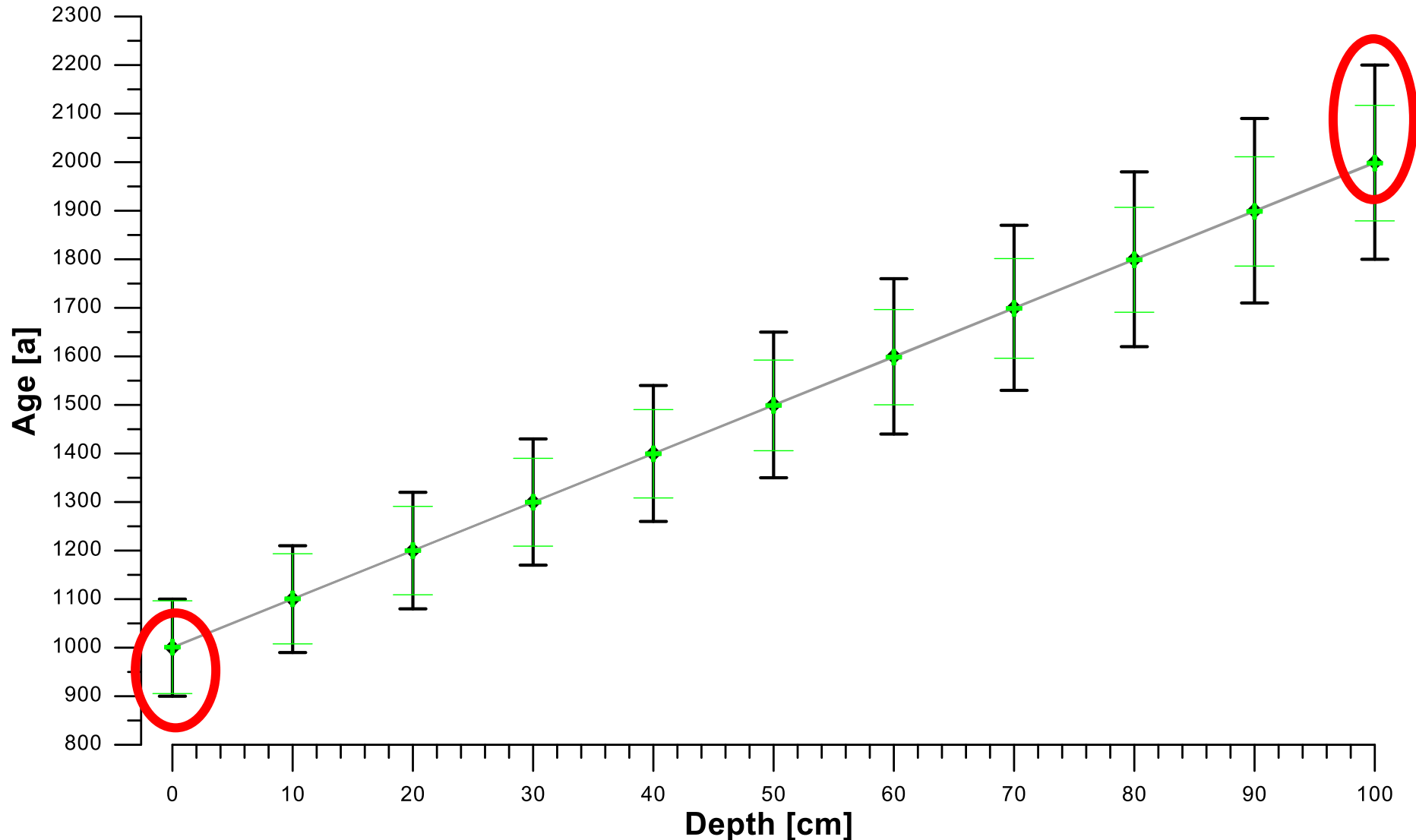
Further software (for various geochronometers)

- **OxCal** (Bronk Ramsey, 1995, 2000, 2009)
- Bacon (Blaauw & Christien, 2011)
- StalAge (Scholz and Hoffmann, 2011)
- MOD-AGE (Hercman and Pawlak, 2012)
- ‚classical code‘ (Blaauw 2010)
- Chronomodel (Lanos and Philippe 2017)
- Archaeochron (Philippe & Vibet 2017)

#1: Bacon: assumes mean SR



Test ages (black) vs. Bacon results (green)



Challenges & Look-out

- Systematic and random uncertainty
- Extracting dates for non-dated stratigraphic levels: How?
- Rates from dates (with their uncertainty)
- Age uncertainty → proxy uncertainty
- Apply, test, learn – feel free to ask

Take home

- Document in detail what you do
- Outer ages: the same before and after modelling?
- Separate random/systematic uncertainty!
Only use random part in models,
recombine afterwards
- Treat modeled ages for undated layers
with caution
- If in doubt: ask an expert

**We hope you learned something
and you liked the course**

Critique, feedback, questions:
Please feel free to contact us.